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MIMO versus Decentralized Control of the Coupled Droop-Share and Drive Loops on the SC001 Balancer Board: An Internal Comparison

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Abstract: The SC001 scale car carries two independently designed robust loops: an H_∞/Youla-H current-share controller that commands a droop ratio between a fuel cell and a battery, and a vehicle-speed drive controller. Both were designed single-input single-output, on the assumption that the two loops do not interact. This report tests that assumption. We build a 2×2 small-signal design plant in which the drive channel disturbs the share channel through the bus current it draws, validate it against an independent 15-state converter-level model, synthesize a centralized 2×2 H_∞ controller with a matrix generalization of the Youla-H DC-correction recipe that makes $T(0) = I$ structural, and compare it head-to-head against the decentralized pair on the same coupled plant, the same corner families and the same simulations. The coupling is one-directional, so the relative gain array is provably uninformative; the informative measure is the directional ratio $|G_{s,12}|/|G_{s,11}|$ on the error- and actuator-normalized plant, which is 1.57 at the nominal 2 A load and 25.1 at 0.5 A. The comparison gives three different answers on the three primary metrics: the centralized controller outperforms the decentralized pair on worst-corner sensitivity by 27 %, ties on regen bus excursion, and is superior or inferior on drive-transient share excursion depending on the sign of an uncalibrated source-voltage mismatch ΔV_0. Decentralized control is stable at all 4992 feasible corners, retains a 2.3× drive-cycle tracking advantage, and does not depend on that sign. We conclude that decentralization is justified on its merits, and that measuring ΔV_0 on the bench is the single highest-value action for narrowing the remaining uncertainty. Methodologically, the actuator limit functions as a design node: it sets the input scaling, hence the effective control-effort penalty, and hence the achievable robustness.

Keywords: H_∞ control; Youla parameterization; MIMO control; droop control; power sharing; decentralized control; hybrid vehicle; DC bus

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1. Introduction

The SC001 scale car's DC balancer board routes power from a fuel cell and a battery onto a shared bus through two off-the-shelf boost regulators whose output resistance is made electronically programmable by injecting a shunt-derived, multiplying-DAC-scaled signal into each converter's feedback node. The resulting *droop ratio r* is the actuator of a current-share loop whose output is the share fraction $\alpha = I_{FC}/(I_{FC} + I_{BT})$, closed by a robust H_∞/Youla-H controller synthesized against a 60-corner uncertain plant family and documented separately [9]. The vehicle's speed is regulated by a second loop that commands motor current through a VESC over UART, following the drive-controller recipe of [5] with the post-synthesis DC correction of [6].

Both loops were designed single-input single-output, each on its own plant, each validated only against its own uncertainty family. That separation was assumed without derivation: the two loops share a bus, and every ampere the motor draws is an ampere the two sources must split. This report tests whether the decentralized design is genuinely justified or merely convenient. Three metrics decide it — worst-corner $\|S\|_\infty$, drive-transient share excursion, and regen-event bus excursion [9].

Two safeguards keep the comparison from being decided by its own construction. *Both* controllers are closed against the same coupled 2×2 plant with the same assembly routine, corner sets and simulator, so the decentralized pair is never evaluated on a diagonal plant; and the amplitude of every excitation is stated, with every case in which the ± 20 A motor-current clamp is active flagged.

1.1. Contributions

- A 2×2 small-signal design plant for the coupled share/drive problem (Section 2), in which the drive-to-share coupling is *derived* from the static share law instead of asserted, and is reproduced endogenously to 0.032 % by an independent 15-state converter-level model (Appendix A).
- The observation that the plant is structurally upper-triangular, so the relative gain array is the identity at every frequency and every corner and carries no information; the informative measure is the directional ratio $|G_{s,12}|/|G_{s,11}|$.
- A matrix generalization of the scalar Youla gain-tuning recipe that enforces $T(0) = I$ on a MIMO loop by a constant right multiplier on the Youla parameter, made *structural* by an exact rank-2 integrator split (Section 4.2).
- A head-to-head comparison on identical terms (Section 6), whose result is that the three primary metrics give three different answers, backed by an independent MATLAB cross-validation of the whole chain (Appendix E).

2. The Coupled 2×2 Plant

2.1. Notation

The share literature, the powertrain-Youla papers and the firmware reuse the same letters for different quantities. Table 1 is binding here; the three collisions that matter in practice are r (droop ratio, never tire radius — that is r_t), α (current share, never an acceleration) and K (share-plant DC gain, never a controller gain or motor K_V).

Table 1. Notation. Small-signal deviations carry a leading Δ.

Symbol	Units	Meaning (and what it is *not*)
r	—	Droop ratio command, $r \in [0.15, 0.85]$. *Not* tire radius.
r_t	m	Tire radius, 0.033.
α	—	Current share $I_{FC}/(I_{FC}+I_{BT})$; the share-loop output. *Never* an acceleration.
k_d	Ω	Droop scale, 0.30 (firmware `K_DROOP`).
$R_{e,max}$	Ω	Full-scale programmable output resistance, 2.014 (as-built).
ΔV_0	V	No-load source-voltage mismatch, FC vs. BT; ± 0.4 budget, sign unknown.
I_{tot}	A	Total bus current out of the two sources.
K	—	Share-plant DC gain $\partial\alpha/\partial r$ at the operating point.
K_v	—	Drive-channel structural gain uncertainty multiplier, $\in \{0.5, 1, 2\}$.
k_t, φ, b_{eff}	—	Motor torque constant, drive ratio 9.49:1, wheel-referred damping.
G, G_s	—	Design plant; $G_s = D_e^{-1} G D_u$ its scaled form.
S, T, Y; W_p, W_Δ, W_u	—	Sensitivity, complementary sensitivity, Youla transfer function (2×2), and the H_∞ cost weights on each (W_Δ, also written W_3, is the robustness weight on T).
M_{YH}, γ	—	MIMO Youla-H DC correction matrix $[G_s(0)Y_H(0)]^{-1}$; H_∞ level.
M_2	—	$M_2 = 1/\|S\|_\infty$, the M_2 stability margin (Def. 4.2 of [3]).
P_c, P_f	—	DGKF control and filter Riccati solutions. *Not* X/Y — Y is the Youla transfer function.

The symbols S, T, Y, L, G_p and G_c follow [3] (pp. xxxi–xxxii, 88–91, 393–395), including that body's definition of $Y = G_c S$ as the *Youla transfer function (matrix)* — which is why Y is not available here as a Riccati solution symbol.

The plant inputs and outputs are

$$u = \begin{bmatrix} \Delta r \\ \Delta i_{cmd} \end{bmatrix}, \qquad y = \begin{bmatrix} \Delta\alpha \\ \Delta v \end{bmatrix}, \tag{1}$$

with actuator limits $r \in [0.15, 0.85]$ and $|i_{cmd}| \leq 20$ A. Native rates are 1 kHz for the share loop and 500 Hz for the motor loop, the latter set by the VESC UART frame floor of roughly 781 µs.

2.1.1. The actuator limit

The motor-current clamp is a *motor-side* limit, but the constraint it must respect lives on the *bus*: the two converters can supply roughly 67 W to 87 W between them. The two are related by the bus-current conversion $A_i \approx 0.243\,\mathrm{A\,A^{-1}}$ of (6), so a ± 20 A motor-side clamp is ≈ 4.9 A bus-side at cruise, ≈ 77 W at $V_{bus0} = 15.9$ V — inside that budget, and close enough to it that the motor clamp and the converter power budget bind *together* by construction. The bus-power limit is therefore not carried as a separate modelled constraint: at the cruise operating point it is approximately equivalent to the motor clamp, and away from that point it is the motor clamp that is quoted. A clamped case is therefore one in which the powertrain as a whole is at its limit, not one in which an arbitrarily chosen number is.

2.2. Share channel

The droop hardware realizes a commanded output resistance $R_e(g) = R_{e,max}\, g$ per channel; mapping the ratio as $R_F = k_d/r$, $R_B = k_d/(1-r)$ gives $R_F \parallel R_B = k_d$ independently of r, so the bus droop is decoupled from the split. Solving the two-Thévenin circuit yields the exact static share law [9]

$$\alpha(r, I_{tot}) \;=\; r \;+\; \frac{\Delta V_0\, r(1-r)}{k_d\, I_{tot}}, \tag{2}$$

whose r-derivative is the share-plant DC gain

$$K \;=\; \frac{\partial \alpha}{\partial r} \;=\; 1 + \frac{\Delta V_0 (1-2r_0)}{k_d\, I_{tot0}}. \tag{3}$$

The share channel of the design plant is then the SISO share plant of [9] unchanged,

$$G_{11}(s) = K \cdot \text{Padé}_2\!\left(e^{-T_d s}\right) \cdot \frac{1}{\tau_r s + 1} \cdot \frac{1}{\tau_f s + 1}, \tag{4}$$

with $T_d = 1\,\text{ms}$ nominal loop delay, $\tau_r = 100\,\mu\text{s}$, and $\tau_f = 0.8\,\text{ms}$ the firmware's 200 Hz share-measurement prefilter; it reproduces the SISO plant to 8.0×10^{-16} relative over all operating points and five parameter corners.

2.3. Drive channel

The drive channel is a VESC transport delay and current-loop lag feeding a force constant and a first-order longitudinal mode:

$$G_{22}(s): \quad \Delta i_{cmd} \to \text{Padé}_2\!\left(e^{-T_{d,v} s}\right) \to \frac{1}{\tau_v s + 1} \to \frac{k_t \eta_{dt} \varphi}{r_t}\,[\text{N}\,\text{A}^{-1}] \to \frac{1}{m_{eff} s + b_{eff}} \to K_{enc} K_v, \tag{5}$$

with DC gain $G_{22}(0) = K_{enc} K_v k_t \eta_{dt} \varphi / (r_t b_{eff}) = 3.7085\,\text{m}\,\text{s}^{-1}\,\text{A}^{-1}$ and a mechanical pole at $-b_{eff}/m_{eff} = -0.1219\,\text{rad}\,\text{s}^{-1}$ — a near-integrator, an 8 s time constant. This recovers the powertrain form of [5] exactly when b is written shaft-referred; here it is written wheel-referred, with the motor's spinning loss reflected in through $b_{motor}(\varphi/r_t)^2$.

Two features of b_{eff} are counter-intuitive. At the nominal operating point aero contributes $0.0245\,\text{N}\,\text{s}\,\text{m}^{-1}$ and the motor's own free-run loss 0.3352, so the motor dominates the small-signal damping by 14×. And rolling resistance is Coulomb: it sets the operating-point torque (hence the coupling gains below) but its derivative with respect to v is zero away from standstill, so it must *not* appear in b_{eff}. The gain uncertainty $K_v \in \{0.5, 1, 2\}$ is structural in origin: the encoder sits downstream of the 9.49:1 reduction *and* the limited-slip differentials, so there is no fixed rotor-to-ground-speed mapping, and k_t is uncalibrated because no motor has been selected.

2.4. Coupling

$G_{21} = 0$ is a modelling *result*: the droop ratio reaches no state that feeds the speed output in either the design plant or the truth model, because the VESC runs a current loop: at fixed i_{cmd} the delivered torque is independent of v_{bus} until the duty rail. The residual this discards is measured — $|\Delta v_{bus}/\Delta r|$ at DC is 0.2002 V per unit r, 0.44 % of the 15.9 V bus over the full r span — well below duty saturation throughout the envelope. It would break under deep battery discharge with a large i_{cmd}, or a regen event pushing the bus into overvoltage protection; both lie outside the linear envelope and are handled by the firmware's sequencing layer.

The live coupling is G_{12}, drive to share, in two stages. Linearizing motor input power $P = (k_t \omega i_m + R_m i_m^2)/\eta_v$ about the operating point at fixed V_{bus0} gives the bus-current draw

$$\Delta I_{bus} = A_i \, \Delta i_m + A_\omega \, \Delta\omega, \qquad A_i = \frac{k_t \omega_0 + 2R_m i_{m0}}{\eta_v V_{bus0}}, \qquad A_\omega = \frac{k_t i_{m0}}{\eta_v V_{bus0}}, \tag{6}$$

numerically $A_i = 0.242\,92\ \mathrm{A\,A^{-1}}$ and $A_\omega = 3.9272 \times 10^{-4}\ \mathrm{A\,rad^{-1}\,s^{-1}}$ at nominal. The $\Delta\omega$ term is what makes the coupling *dynamic* rather than a static feedthrough: it routes through the vehicle mode, so the coupling inherits the near-integrator pole. Differentiating (2) with respect to total current gives the share sensitivity

$$\frac{\partial \alpha}{\partial I_{tot}} = -\frac{\Delta V_0 \, r_0 (1 - r_0)}{k_d \, I_{tot0}^2}, \tag{7}$$

so that $G_{12}(s) = (\partial\alpha/\partial I_{tot}) \left[A_i W_v(s) + A_\omega (\varphi/r_t) \frac{1}{m_{\text{eff}} s + b_{\text{eff}}} W_v(s) \right] / (\tau_f s + 1)$ with W_v the VESC path and $1/(m_{\text{eff}} s + b_{\text{eff}})$ the vehicle mode. At nominal $\partial\alpha/\partial I_{tot} = -4.1667 \times 10^{-2}$ share per ampere and $G_{12}(0) = -2.7573 \times 10^{-2}$ share per ampere of i_{cmd}.

Both G_{11} and G_{12} pass through the *same* instance of the prefilter $1/(\tau_f s + 1)$, because there is exactly one such filter in firmware, downstream of the share estimate; the plant is therefore assembled as explicit block state-space instead of a block-diagonal composition, since duplicating the filter would add a spurious state and misrepresent the hardware. The nominal assembly is 8 states.

2.4.1. Sign uncertainty of the coupling

Equation (7) is *exactly zero* at $\Delta V_0 = 0$ (matched sources: no coupling) and *flips sign* over the ± 0.4 V budget, which is an uncalibrated quantity. A centralized controller synthesized at one sign of ΔV_0 implements a coupling feedforward of the wrong sign at the mirror corner. Both signs are therefore mandatory in every evaluation below.

2.5. Why the RGA is the wrong instrument

The design plant is upper triangular, so its relative gain array is the identity — $\max|\mathrm{RGA} - I| = 2.23 \times 10^{-16}$ across all frequencies and all cases, including $\Delta V_0 = \pm 0.4$. This is a property of *triangularity*: $\mathrm{RGA}(T) = I$ for any triangular T no matter how large the off-diagonal entry. The RGA is blind to one-way coupling and must not be used as the coupling metric here.

The informative measures, computed on the scaled plant, are the directional ratio $|G_{s,12}|/|G_{s,11}|$ and the condition number ([3] §12.7, p. 414)[1] (Table 2, Figure 1). Both are computed on the scaled plant, so both inherit the normalization: a ratio of unity means "a full-authority motor command moves the share as much as a full-authority droop command does." At the nominal 2 A load the drive-to-share path is 1.57× the direct path; at 0.5 A it is 25× it, because (7) scales as $1/I_{tot}^2$ — a full-authority motor transient moves the share by far more than a full-authority droop command can. That figure is the number that motivates a centralized design.

1 That source writes the condition number $K(\cdot)$; this report writes $\mathrm{cond}(\cdot)$ to avoid collision with its several K-symbols (Table 1).

Table 2. Coupling quantification on the scaled plant. The in-band window is DC to 200 rad s^{-1}: $\mathrm{cond}(G_s(j\omega)) \to \infty$ as $\omega \to \infty$ for any strictly proper plant whose channels roll off at different orders, so a full-sweep maximum would measure roll-off asymmetry rather than coupling.

Quantity	**Nominal** (2 A, $r = 0.5$)	**Light load** (0.5 A)	**FC cruise** ($r = 0.85$)
peak $\lvert G_{s,12}\rvert / \lvert G_{s,11}\rvert$	1.5710	**25.136**	1.0451
$\mathrm{cond}(G_s)$ at DC	21.238	51.064	27.588
$\mathrm{cond}(G_s)$, in-band	101.9	6429.5	67.49
$\partial\alpha/\partial I_{tot}$ [share/A]	−0.04167	**−0.66667**	−0.02125
Over the whole feasible operating grid (26 of 30 points feasible):			
worst $\lvert\partial\alpha/\partial I_{tot}\rvert$	**1.120** share/A at 0.5 A, $r_0 = 0.7$, $\Delta V_0 = -0.4$ V		
worst DC $\mathrm{cond}(G_s)$	51.15		
worst in-band $\mathrm{cond}(G_s)$	8834.6		
max $\lvert\mathrm{RGA} - I\rvert$	2.23×10^{-16} — all cases, all frequencies		

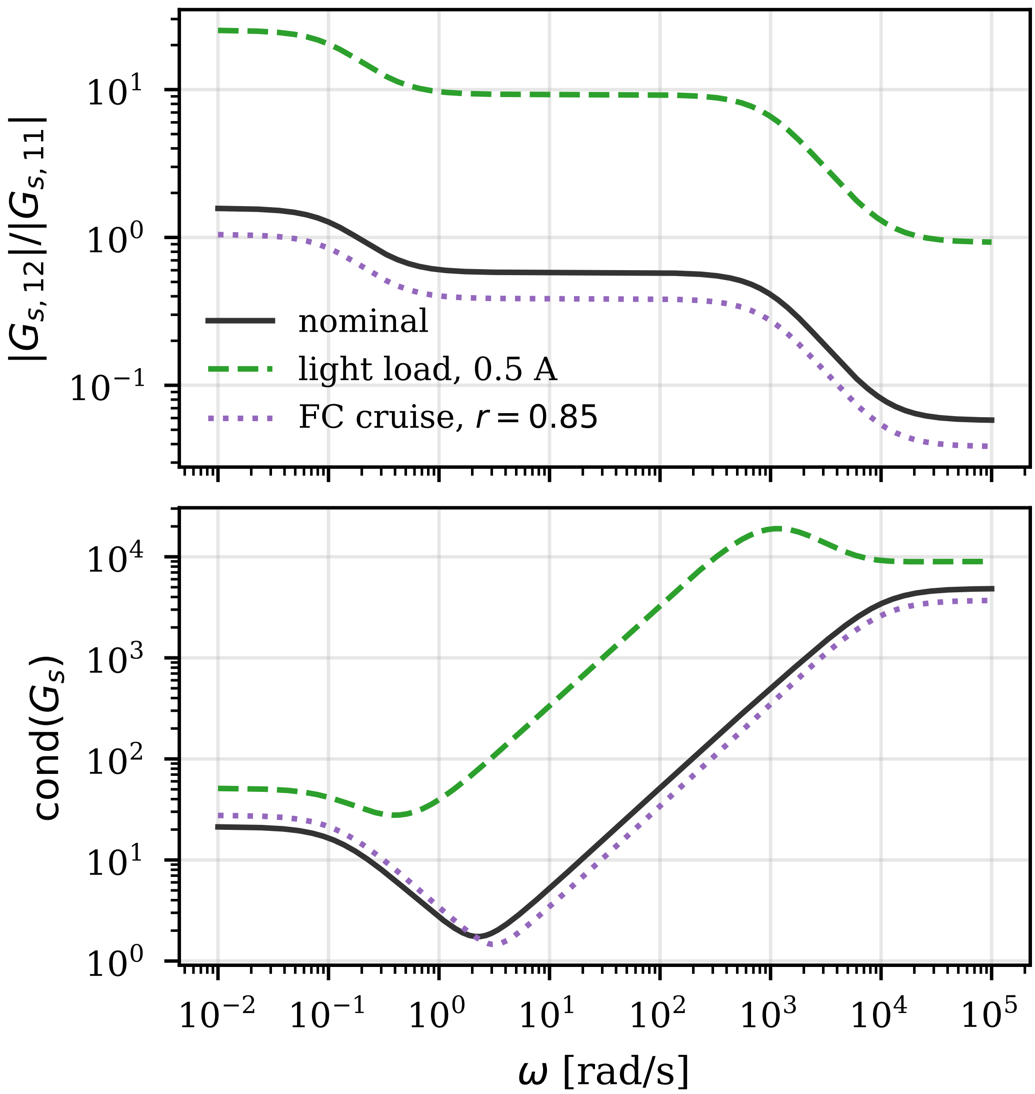

Figure 1. Coupling versus frequency for three scenarios (nominal 2 A $r = 0.5$; light load 0.5 A; FC cruise $r = 0.85$). Top: directional ratio $\lvert G_{s,12}(j\omega)\rvert / \lvert G_{s,11}(j\omega)\rvert$ on the scaled plant. Bottom: $\mathrm{cond}(G_s(j\omega))$ for the same scenarios. All three curves reach or exceed unity in band and the light-load curve exceeds it by more than an order of magnitude: that is the whole motivation for considering a centralized design; the RGA is identically I for all three and would show nothing.

2.6. Operating envelope and validation

An operating point is the 4-tuple OP $= (I_{tot0}, r_0, \Delta V_0, v_0)$, nominally $I_{tot0} = 2\,\mathrm{A}$, $r_0 = 0.50$, $\Delta V_0 = +0.2\,\mathrm{V}$ and $v_0 = 2\,\mathrm{m\,s^{-1}}$; half the $\pm 0.4\,\mathrm{V}$ budget is used for ΔV_0 deliberately, since synthesizing at zero would show the controller no coupling at all and at ± 0.4 would over-commit to one sign. The uncertainty families are 10 operating points, 24 share corners and 24 drive corners; of the $10 \times 24 \times 24 = 5760$ Tier-1 combinations, **768 are infeasible** — the unidirectional ideal-diode switches make (2) meaningless where it puts α_0 outside $(0, 1)$ — and are skipped and counted instead of silently linearized, leaving the **4992 feasible corners** used throughout. Synthesis and all frequency-domain metrics run on the scaled plant $G_s = D_e^{-1} G D_u$ with $D_e = \mathrm{diag}(0.05, 0.5)$ and $D_u = \mathrm{diag}(0.35, 20.0)$, giving $G_s(0) = \left[\begin{smallmatrix} 7.0 & -11.03 \\ 0 & 148.3 \end{smallmatrix}\right]$ and $\mathrm{cond}(G_s(0)) = 21.31$. Writing K as (3) rather than sweeping it makes the SISO envelope $K \in [0.55, 1.45]$ *derived*, and exposes a gap: over the full feasible operating set the implied envelope is $[0.533, 2.067]$, 43 % above it at light load with full mismatch. Those corners are carried as stability-only corners. Full corner-family definitions, feasibility arithmetic and envelope tables are in Appendix D. The design plant is independently checked against a 15-state converter-level truth model in which the coupling *emerges from the circuit* rather than being written in, agreeing to 0.032 % at DC and within 0.93 % in band at nominal (Appendix A).

3. The Two SISO Designs

3.1. Share controller

The share half of the decentralized baseline is the existing firmware share controller [9]: an H_∞/Youla-H design on the SISO share plant (4) with $\gamma_{opt} = 0.6532$, built at a 5 % back-off ($\|T_{zw}\|_\infty = 0.6841$) and realized as three biquads plus a trapezoidal integrator at 1 kHz with back-calculation anti-windup. It achieves $T(0) = 1$ exactly by the scalar Youla-H correction, an S/T crossover near $110\,\mathrm{rad\,s^{-1}}$, a nominal M_2 margin of 0.80 ([3], Def. 4.2 and Props. 4.2–4.3, pp. 148–149) with 73.9° phase margin, and a 11.7 ms delay margin — roughly six times the worst modeled delay. Against a 60-corner family it is stable everywhere with worst-corner discrete $\|S\|_\infty = 1.87$, against 26.9 for the nominally tuned PI it replaced. It is used here *unchanged*.

3.2. New SISO drive controller

The drive half of the baseline did not exist and was built for this study, following the recipe of [5,6] on the drive plant (5): a second-order strictly-proper weight on S, a `makeweight` roll-off weight on T, and a control-effort weight on Y. Two deviations were forced. The papers' W_Δ break at $18\,\mathrm{rad\,s^{-1}}$ sits *below* the W_p corner on this plant, making the two weights contend over the same decade, inflating γ and collapsing the achieved crossover; the break was moved to 2.5× the W_p corner, preserving the intent. Second, the achieved crossover lands consistently *below* the W_p corner with $\gamma_{opt} \gg 1$ throughout: **the specifications are being shaped, not met**. The cause is that a second-order strictly-proper S weight demands two integrators in the loop below its corner, and the drive plant's mechanical pole at $0.122\,\mathrm{rad\,s^{-1}}$ contributes none of them, leaving only the controller's single integrator; no reweighting brings γ below about 1.2 (Appendix B). W_p was therefore cornered at $50\,\mathrm{rad\,s^{-1}}$ to land the achieved crossover near the papers' $24\,\mathrm{rad\,s^{-1}}$, at $\gamma_{opt} = 13.03$.

A γ of 13.03 means the weights are not met; the controller itself is sound. The delivered loop has $\|S\|_\infty = 1.3851$ nominal, 48.9° phase margin at $20.75\,\mathrm{rad\,s^{-1}}$, 18.3 dB gain margin and a 41.12 ms delay margin, and is stable at all 72 drive corners with worst continuous $\|S\|_\infty = 2.4074$. The scalar Youla-H correction is tiny here ($T_H(0) = 0.99864$, a gain scale of 1.001 37) and delivers $T(0) = 1.000000000000$.

Anti-windup drives the cost comparison of Section 5. The reduced controller's *non-integral* branch alone has a DC gain of $367.7\,\mathrm{A\,m^{-1}\,s^{-1}}$, so it saturates the ± 20 A actuator for speed errors above $e_{sat} = 54.4\,\mathrm{mm\,s^{-1}}$ and its lag states wind up independently of the integrator. The failure is therefore *amplitude-bounded*. A step-amplitude scan of the integrator-only back-calculation scheme (pass: final error below $10^{-4}\,\mathrm{m\,s^{-1}}$ and tail peak-to-peak below $10^{-5}\,\mathrm{m\,s^{-1}}$) is clean at 0.02, 0.05, 0.10 and $0.20\,\mathrm{m\,s^{-1}}$ — final errors of order $10^{-8}\,\mathrm{m\,s^{-1}}$ — and fails from $0.5\,\mathrm{m\,s^{-1}}$ upward, reaching a $-0.101\,\mathrm{m\,s^{-1}}$ standing error and a $6.3\,\mathrm{mm\,s^{-1}}$ limit cycle on the 0 to $2\,\mathrm{m\,s^{-1}}$ step. **Integrator-only conditioning is adequate for small transients and inadequate for large ones**; since the 0 to $2\,\mathrm{m\,s^{-1}}$ step is a required gate, the baseline still *requires* Hanus self-conditioning [7] on a five-state state-space realization (Appendix F), which is *not* implementable as a biquad cascade.

4. Centralized 2×2 H_∞ / Youla-H Synthesis

4.1. Weights and the DGKF solution

Synthesis uses block-diagonal weights on the three-block stack $[\,W_pS;\ W_uY;\ W_\Delta T\,]$, one philosophy per channel (Table 3). This is the *loop shaping* problem in the sense of [3] §6.4, eqs. (6.44)–(6.48), pp. 229–232 (SISO) and §14.5, eq. (14.67) and Fig. 14.20, p. 468 (MIMO), whose Youla-based formulation is the method this report follows; [1,2] are complementary references for the same machinery.[2] The share block is the SISO set of Section 3, unchanged. The drive block is *not* the Section 3 set — it was iterated independently against the MIMO corner battery (Appendix B) — and that difference bounds how the two designs may be compared (Section 6.1). The binding constraint on this plant turns out to be corner robustness instead of nominal performance, so γ **is the wrong scalar to minimize here**: the final drive block deliberately accepts a worse γ than an available alternative in exchange for corner margin, paying about 60 ms of small-signal drive settling.

Table 3. Final block-diagonal weights. Bandwidth separation (share $\approx 110\,\mathrm{rad\,s^{-1}}$ vs. drive $24\,\mathrm{rad\,s^{-1}}$) is deliberate: the share loop rides on 1 ms electrical dynamics, the drive loop on an 8 s mechanical mode. All γ values are a-posteriori achieved levels.

Block	**Share channel**	**Drive channel**
W_p (on S)	`strictly_proper_lf`$(10^4, 40)$	`strictly_proper_lf`$(10^4, 24)$
W_Δ (on T)	`makeweight`$(0.5, 250, 40)$	`makeweight`$(0.5, 60, 40)$
W_u (on Y)	`makeweight`$(0.3, 600, 20)$	`makeweight`$(0.5, 18, 20)$
γ: share alone / drive alone / MIMO		0.5760 / 1.6429 / **1.6456**

The augmented plant has 14 states with $p_z = 6$, $p_y = 2$, $m_u = 2$, and is solved by the two-Riccati DGKF construction [8] (equivalently [3] §15.3, Thm. 15.2, eqs. (15.94)–(15.98), pp. 491–493). The same source's Remark on p. 492 — that a control input carrying no weight breaks the D_{12} rank condition — is why W_u is non-empty on both channels: it is a solvability requirement, not a tuning preference. Two features of the solution bear on every γ in this report. First, the measurement-equals-reference structure gives $D_{21} = I_2$, which makes the filter Riccati degenerate with $P_f \equiv 0$ as the provably stabilizing solution — reproduced to 1.0×10^{-14}, which serves as a correctness check on the estimation half of the solver at no additional cost. Second, because $P_f \equiv 0$ the DGKF feasibility test $\rho(P_cP_f) < \gamma^2$ is vacuous, so the Riccati bisection is optimistic in MIMO and must not be gated on: it reports $\gamma = 0.89769$ while the central controller attains $\gamma_{achieved} = \mathbf{1.64563}$ with

[2] The wider literature often calls this three-block form "mixed-sensitivity $S/KS/T$". We follow [3], which reserves *mixed sensitivity* for the two-block S/T problem (§6.6, §14.5) and calls the three-block form loop shaping.

$\|T_{zw}\|_\infty = 1.64573$. **Every γ quoted in this report is an a-posteriori achieved level.** Details and formulas are in Appendix C.

Because $G_{21} = 0$, any MIMO controller restricted to $e_1 = 0$ acts as a drive-channel SISO controller, so $\gamma_{MIMO} \geq \gamma_{drive}$ necessarily; measured, $1.6456 \geq 1.6429$, equal to 0.16 %: **the coupling costs essentially nothing**, since the centralized controller attains the drive channel's own unconstrained optimum while also handling G_{12}. And at $\Delta V_0 = 0$ the plant is exactly diagonal, so the problem must decouple and $\gamma_{MIMO} = \max(\gamma_{share}, \gamma_{drive})$; measured, 1.641 87 vs. 1.642 95 — 0.07 % on a case with a known closed-form answer.

4.2. The MIMO Youla-H DC correction: $T(0) = I$

This is the report's control-theoretic contribution. $T(0) = I$ is the MIMO statement of perfect low-frequency tracking, zero steady-state error and input–output decoupling ([3] §12.7.1, p. 414), recognizable by $\bar{\sigma}(T) = \sigma_{\min}(T) = 1$ at low frequency; its SISO ancestor is the interpolation condition $G_p(0)Y(0) = T(0) = 1$ of §3.4, eq. (3.64), p. 113, which the matrix correction below generalizes. The Youla route of [3] enforces $T(0) = 1$ *by construction*, as an interpolation condition imposed while Y is shaped (§3.5); H_∞ synthesis honours no interpolation conditions natively, so it delivers a controller that is only approximately integral. The correction here therefore restores post-synthesis a property that the book's own route obtains by construction — it is a repair, not a new capability. For a SISO plant G_P, the recipe of [6] takes the H_∞-synthesized Youla parameter in zero-pole-gain form,

$$Y_H(s) = K_H \frac{\prod_{i=1}^{n}(s+z_i)}{\prod_{j=1}^{m}(s+p_j)}, \tag{8}$$

keeps every z_i and p_j, and rescales the gain alone to $K_{YH} = K_H/(Y_H(0)G_P(0))$, which enforces $T_{YH}(0) = G_P(0)Y_{YH}(0) = 1$ and, since $G_C = Y/S$ and $S(0) = 0$, plants at least one pure integrator in the realized controller.

A scalar gain has no matrix analogue that preserves "the same zeros and poles." But it has one that preserves the *dynamics*: a constant *right* multiplier. Build the Youla parameter by state-space interconnection, $Y_H = K_H(I + G_s K_H)^{-1}$, valid because $D_K = 0$, so that $T = G_s Y_H$ exactly as in the scalar case. Then define

$$M_{YH} \;:=\; [G_s(0)\, Y_H(0)]^{-1} \in \mathbb{R}^{2\times 2}, \qquad Y_{YH}(s) \;:=\; Y_H(s)\, M_{YH}. \tag{9}$$

Since M_{YH} multiplies only the input matrix of the state-space realization, the poles and zero dynamics of Y_H are untouched — the exact matrix analogue of "keep z_i and p_j, change K." The correction is immediate:

$$T_{YH}(0) \;=\; G_s(0)\, Y_{YH}(0) \;=\; G_s(0)\, Y_H(0) [G_s(0) Y_H(0)]^{-1} \;=\; I. \tag{10}$$

The scalar recipe is recovered exactly when the problem is 1×1: $M_{YH} = 1/(G_P(0)Y_H(0))$, i.e. $K_{YH} = K_H/(Y_H(0)G_P(0))$.

The choice of *right* multiplication is forced twice over. Algebraically, $T = G_s Y$, so only a right factor of Y can be inverted against $G_s(0)Y_H(0)$ to give the identity; $M_{YH}\, Y_H$ would give $G_s(0) M_{YH}\, Y_H(0) \neq I$ in general because matrices do not commute. Physically, left-multiplying would mix the controller *outputs* rather than re-map its inputs, changing which actuator answers which error — yielding a different controller instead of a corrected one.

The realization follows the scalar case block-for-block: $G_{c,YH} = Y_{YH}(I - G_s Y_{YH})^{-1}$, a positive feedback of G_s around Y_{YH}, well-posed because $D_Y = 0$. The final step is what makes the property robust rather than merely achieved: an ordered real Schur decomposition plus Sylvester block-diagonalization separates the two near-origin poles of

$G_{c,YH}$ into an exact residue bank K_I/s plus a stable remainder. **After the split,** $T(0) = I$ **is structural** — two exact integrators with a non-singular K_I force $S(0) = 0$ regardless of any subsequent rounding, truncation or float32 quantization, so the DC property no longer depends on a numerically delicate pole-at-almost-zero, which is the risk the scalar recipe carries into a fixed-point implementation. Numerically,

$$K_I = \begin{bmatrix} 7.992467 & 0.1645500 \\ 7.8 \times 10^{-5} & 0.1031860 \end{bmatrix}, \tag{11}$$

nearly lower triangular with a small $(1,2)$ entry and a negligible $(2,1)$ entry: the DC coupling correction runs drive → share, matching the plant's triangularity.

The correction is a small adjustment rather than a substantial repair. Here $\mathrm{cond}(G_s(0)Y_H(0)) = 1.0001$ — the book's low-frequency decoupling recognizer $K(T_y) \approx 1$ ([3] §12.8.1, p. 416) — and $\|M_{YH} - I\|_2 = 1.57 \times 10^{-4}$, saying the H_∞ controller was already integral to four decimals, as one expects from an S weight with DC gain 10^4 (a large $\|M - I\|$ would have implicated the weights themselves). The delivered $\|T(0) - I\|_{\max}$ is 6.5×10^{-10}, preserved through reduction. The correction's remaining job is therefore not to *create* the integrator — H_∞ already put a near-integrator there — but to fix its *direction*.

4.2.1. Relation to the direct MIMO Youla design route

The plant here is stable and square, so its Smith–McMillan structure imposes no unstable pole/zero interpolation constraints and the direct MIMO Youla design of [3] §13.2 (pp. 422–425) applies verbatim, with the controller recovered as $Y = G_p^{-1} T_{yd}$ and $G_c = Y S_y^{-1}$ (§13.5, p. 445). It was not used here because the binding constraint on this problem is robustness across the corner family rather than nominal shaping: H_∞ synthesis optimizes a worst-case norm directly, whereas the direct route would require the designer to guess a T_{yd} that survives all 4992 corners. The Youla-H correction above then re-imposes the one property the direct route would have given by construction.

4.3. Utilization of the cross-coupling by the centralized controller

Table 4 quantifies the extent to which the centralized controller uses the cross-coupling.

Table 4. Peak scaled magnitude of each controller entry, and its ratio to the diagonal term of the same row.

Entry	Peak $\lvert\cdot\rvert$ (scaled)	Ratio to its diagonal
K_{11} (share ← share error)	79.92	—
K_{12} (share ← speed error)	**1.7167**	**2.15 %**
K_{21} (drive ← share error)	9.93×10^{-4}	0.074 %
K_{22} (drive ← speed error)	1.3345	—

$K_{21} \approx 0$ is correct and expected: $G_{21} = 0$, so there is nothing the drive actuator can do about a share error, and its 7.4×10^{-4} ratio is numerical residue — a check that the synthesis recovers the plant's triangularity. K_{12} at 2.15 % of K_{11} is the real coupling feedforward: the share actuator pre-empts the bus-current disturbance a commanded acceleration is about to create. It is small in *ratio* because K_{11} must be large (the share loop's 110 rad s^{-1} bandwidth); the term itself is fully active. Its magnitude is bounded by the ΔV_0 sign uncertainty by construction: a feedforward built at +0.2 V is the wrong sign at −0.4 V, and the synthesis assigns it little authority.

The centralized controller is about 98 % **decentralized by gain.** Decentralized control here is therefore justified on its merits, on the centralized design's own evidence.

5. Implementation

The order-30 controller $G_{c,YH}$ reduces, after the exact integrator split and balanced truncation at a 10^{-2} margin, to **9 continuous states**: two exact integrators plus a seven-state remainder, with a maximum relative $\bar{\sigma}$ deviation of 1.88×10^{-3} over $\omega \in [0.1, 10^4]$ against a 2×10^{-2} tolerance. The order is set by the Hankel spectrum: the singular-value tail is flat here (states 6 and 7 carry 9.2×10^{-4} and 6.3×10^{-4} against a leading 0.45), so truncating at five would not clear the 10^{-2} auto-selection margin. **The selection tolerance was deliberately not loosened to force a smaller controller**: two float32 states cost 8 bytes and about 26 MAC per tick, whereas a corner result that survives only because truncation error happened to damp the worst corner is not a design property at all (Appendix B). The whole controller runs single-rate at 2 ms **(**500 Hz**)**, because it issues the motor current command and the VESC UART frame floor caps that channel there; the cost to the share channel is bounded and checked explicitly, since Nyquist at 1571 rad s^{-1} is still 14× the 110 rad s^{-1} share crossover — which is what discharges the standing assumption that discretization has no performance effect at a reasonable sampling rate ([3], p. 261) — and the extra effective delay of at most 1 ms sits inside the SISO design's 11.7 ms delay margin. The entire Tier-1 family is re-verified in discrete time (0 of 4992 unstable). Discretization (exact 2×2 Tustin integrator, real-modal remainder) and the matrix-valued anti-windup scheme are detailed in Appendix F.

Table 5. Implementation on the Teensy 4.1 (600 MHz Cortex-M7). The decentralized state count is quoted as implemented: its drive half needs a six-state Hanus self-conditioned realization rather than the five-state controller order, because the conditioning product adds a state's worth of storage and an n^2 term per tick. The MIMO MAC figure counts the state-space core ($Ax + Be$, $Cx + De$, integrator) only, which is the same basis on which the decentralized halves are counted; counting the D_e^{-1}/D_u scaling and the back-calculation arithmetic as well gives 97 MAC per tick (48 500 MAC per second) for the same code. Both accountings are quoted in the design record; only the like-for-like one belongs in a ratio.

Quantity	Decentralized	MIMO
rate	1 kHz (share) + 500 Hz (drive)	500 Hz single-rate
controller states (implemented)	$4 + 6 = \mathbf{10}$	**9**
anti-windup	scalar back-calculation + Hanus	D_u^{-1} back-calculation + clamp
MAC per tick	18 @ 1 kHz, 63 @ 500 Hz	89 @ 500 Hz
MAC per second	**49 500**	**44 500**
ratio MIMO / dec	—	**0.899**
coefficient floats	—	101 (404 B)
float32 replay error (64-step, saturating)	—	5.52×10^{-6} (tol. 5×10^{-4})

The two are comparably inexpensive, with the centralized controller slightly ahead, at 0.899× the decentralized MAC rate and one fewer state. It is ahead at all only because the decentralized baseline *cannot* be a biquad cascade — its drive half needs the Hanus state-space form, whose $A_d - B_d C_d / D_d$ product costs an extra n^2 per tick, and it runs one of its two halves at 1 kHz. A 10 % margin is not an argument for either architecture: **compute cost does not discriminate here**. Both figures are negligible on this core (under 0.1 % of a single issue slot), so **the compute cost is not the cost**. The genuine costs of centralization are that it moves the share loop from 1 kHz to 500 Hz and couples the two control tasks into one scheduling unit; float32 coefficient sensitivity; and a matrix-valued anti-windup scheme that is more code to get right than the scalar one. The UART frame floor is unchanged either way and remains the true bottleneck.

5.1. Actuator rate limiting of the drive channel

With $K_F = k_t\eta_{dt}\varphi/r_t = 1.334\,\mathrm{N\,A^{-1}}$ and $m_{eff} = 2.95\,\mathrm{kg}$, the $\pm 20\,\mathrm{A}$ clamp bounds acceleration at $a_{max} = 9.04\,\mathrm{m\,s^{-2}}$: no reference the loop can be asked to follow is tracked faster than that, whatever the design bandwidth says. The share step (0.05 share) settles to 2 % in 44 ms and the small-signal drive step ($0.05\,\mathrm{m\,s^{-1}}$, peak 1.18 A) in 150 ms, both with exact final values. A $0.5\,\mathrm{m\,s^{-1}}$ step is also linear at this clamp — its 11.79 A peak sits inside the rail — and settles in 150 ms to 0.500 000. The clamp binds from roughly $1\,\mathrm{m\,s^{-1}}$ of step amplitude upward, which is where the large-signal cases of Section 6 live. **The** $24\,\mathrm{rad\,s^{-1}}$ **bandwidth is still a small-signal claim**; it now holds over a usefully wider amplitude range than the anti-windup boundary $e_{sat} = 54.4\,\mathrm{mm\,s^{-1}}$ of the decentralized drive half.

6. Comparison

6.1. Setup and fairness conditions

Three controllers are compared. `dec` is the decentralized baseline, $\mathrm{blkdiag}(K_{share}, K_{drive})$, with the share controller at 1 kHz and the Section 3 drive controller at 500 Hz. `mimo` is the Section 4 controller at 500 Hz. `dec500` is the decentralized pair with its share half re-discretized at 500 Hz, and exists purely to isolate the sample-rate difference as a confound — it changes the share excursion by 0.8 % (large step) and 7.1 % (small step), leaves the regen bus excursion bit-identical and Tier-1 stability at 0/4992, so **every conclusion below survives equalizing the sample rates**.

The essential fairness condition is that **both controllers are closed against the same coupled** 2×2 **plant**, using the same assembly routine, corner sets and simulations; the decentralized controller is never evaluated on a diagonal plant, which would bias the comparison in its favour by construction. Time-domain results use a multirate discrete simulator at a 0.1 ms base step with the plant ZOH-discretized per corner, so `dec` runs its two halves at their native rates; frequency-domain results use continuous physical-coordinate controllers recovered from the emitted float32 headers by exact inverse Tustin plus the analytic K_I/s integrator. All time-domain signals are small-signal deviations about the nominal operating point.

6.1.1. Corner-family verification in place of a small-gain certificate

The robustness certificate of [3] is the small-gain condition $\bar{\sigma}(W_\Delta T_o) < 1$ against a norm-bounded multiplicative perturbation (§14.5, Thm. 14.3, p. 466). This study does not use it. The dominant uncertainties here — the *sign* of ΔV_0, the structural gain multiplier K_v, and the transport delays — are parametric and structured, and wrapping them in an unstructured norm bound large enough to cover the ΔV_0 sign flip would be so conservative that no controller would certify. Robustness is therefore verified by explicit evaluation over the parametric corner family. W_Δ **in this report is a shaping filter, not an identified uncertainty bound**, and **no claim of** $\bar{\sigma}(W_\Delta T_o) < 1$ **is made or implied** by any γ quoted here.

A further limitation: the two designs use different weight *families* on the drive channel (second-order at $50\,\mathrm{rad\,s^{-1}}$ for the SISO baseline, first-order at $24\,\mathrm{rad\,s^{-1}}$ for the MIMO) and their γ values sit on different plant scalings, so γ **is not comparable between them and is never quoted side by side**. Which controller performs better on the real coupled plant is settled by the closed-loop harness regardless.

6.2. *The three primary metrics give three different answers*

Table 6. The three primary metrics and their verdicts. Worst-corner sensitivity is the Tier-2 in-envelope $\bar{\sigma}(S_o)$; excursions are small-signal ($\Delta v_{ref} = +0.05\,\mathrm{m\,s^{-1}}$, zero time on the current rail).

Metric	`dec`	`mimo`	**Verdict**
worst-corner $\|S\|_\infty$ (in-envelope $\bar{\sigma}(S_o)$)	2.5535	**1.8754**	**MIMO superior**, $-27\,\%$ (both on the same representative corner set)
drive-transient share excursion, $\Delta V_0 = +0.4/-0.4$	0.01744 / 0.01744	**0.00667** / 0.01935	**split by ΔV_0 sign** ($-62\,\%$ / $+11\,\%$)
regen-event bus excursion	1.491 12 V	1.474 29 V	**tie** (ratio 0.989)

6.2.1. Metric 1: worst-corner sensitivity

At the nominal plant the centralized controller has $\|S_o\|_\infty = 1.2287$ against 1.3974, and over the Tier-2 in-envelope family its worst $\bar{\sigma}(S_o)$ is 1.8754 against 2.5535, better at **167 of 168 corners** (Figure 2; full tables and the per-corner scatter in Appendix D). Two qualifications attach to the absolute numbers. The Tier-2 figure is computed on a representative corner subsample, so it is a lower bound on the full-grid worst and an absolute "< 2.0" target read off it is subsample-dependent — but the decentralized comparison point comes from the same set, so the *relative* 27 % verdict stands. And the 195× nominal improvement in the cross-transfer $\|T_{\alpha \leftarrow v_{ref}}\|_\infty$ (0.256 76 against 1.314×10^{-3}) is a continuous-domain figure: on the as-implemented 2 ms loop it is 1.109×10^{-2}, about **23**× better than decentralized. Both controllers are stable at 0 of 4992 Tier-1 corners unstable, in continuous time and in the exact multirate discrete loop — **no decentralized instability was found anywhere**.

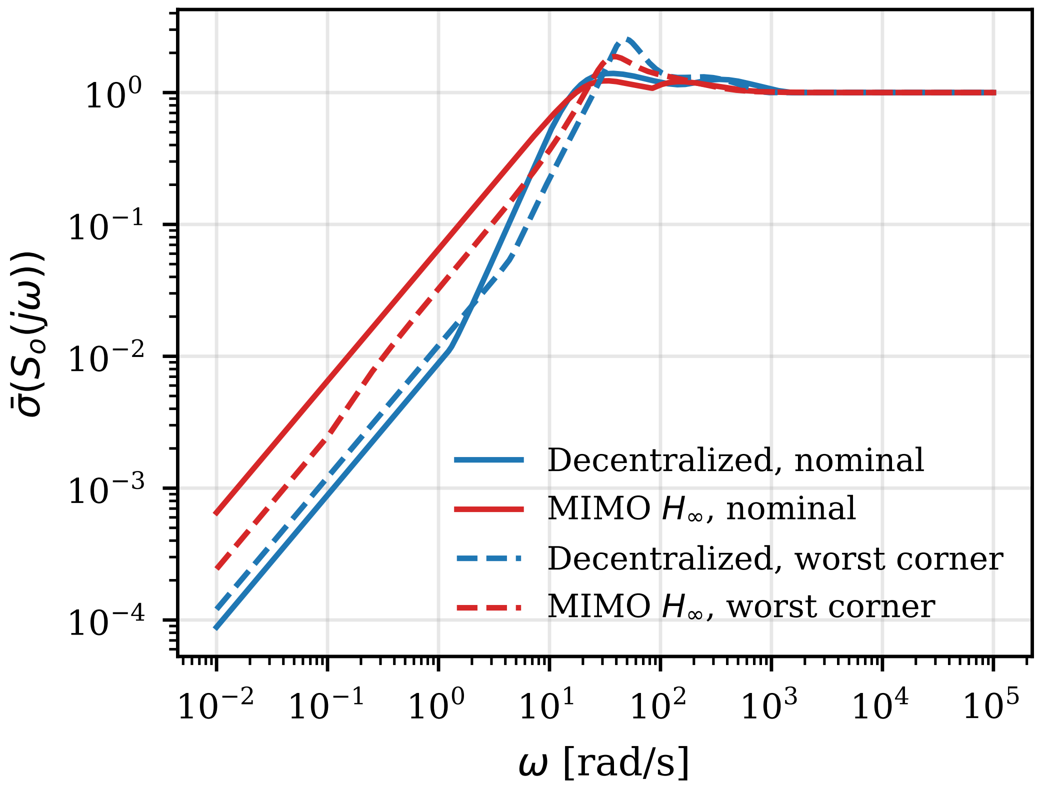

Figure 2. $\bar{\sigma}(S_o(j\omega))$ for both controllers, at the nominal operating point and at the worst in-envelope corner ($I_{tot0} = 2$ A, $r_0 = 0.3$, $\Delta V_0 = -0.4$ V, with the drive channel's gain and delay extremes). Blue is decentralized, red is MIMO H_∞.

6.2.2. Metric 2: drive-transient share excursion

At the sign of ΔV_0 the centralized controller was synthesized for, it cuts the drive-induced share excursion by 62 % and settles faster (0.175 s against 0.189 s) with a 43 % smaller current peak; at the mirrored sign it makes the excursion 11 % worse (Figure 3) — a downside that is now marginal rather than punitive. The decentralized traces are identical between the two signs — as they must be, since a diagonal controller on a linear plant sees only the magnitude of the coupling and is blind to its sign. Splitting the corner statistics by sign over the 96 coupled in-envelope corners, the median cross-transfer ratio MIMO/dec is **0.941** when the sign matches the synthesis and **2.162** when it opposes, spread 0.1938 to 4.378; the centralized controller has the smaller cross-transfer at 28 of 96 corners. **The $\approx 2.3\times$ asymmetry between the matching and opposing medians is insensitive to the actuator span and the operating point**, where the magnitude of the benefit is not. The saturated large step ($+2\,\mathrm{m\,s^{-1}}$) splits by sign in the same way — the centralized controller is 48 % better at the matching sign and 90 % worse at the opposing one — and the decentralized pair reaches a droop clamp ($r = 0.85$ at $\Delta V_0 = +0.4$, $r = 0.15$ at -0.4) at both signs, whereas the centralized one reaches a clamp only at the matching sign (Appendix D). What the decentralized pair keeps outright there is *speed* recovery: it settles $45\times$ faster after the saturation episode, because the centralized drive integrator residue is only $K_{I,22} = 0.103$ against a $0.122\,\mathrm{rad\,s^{-1}}$ vehicle pole, so post-saturation recovery is slow *by design*.

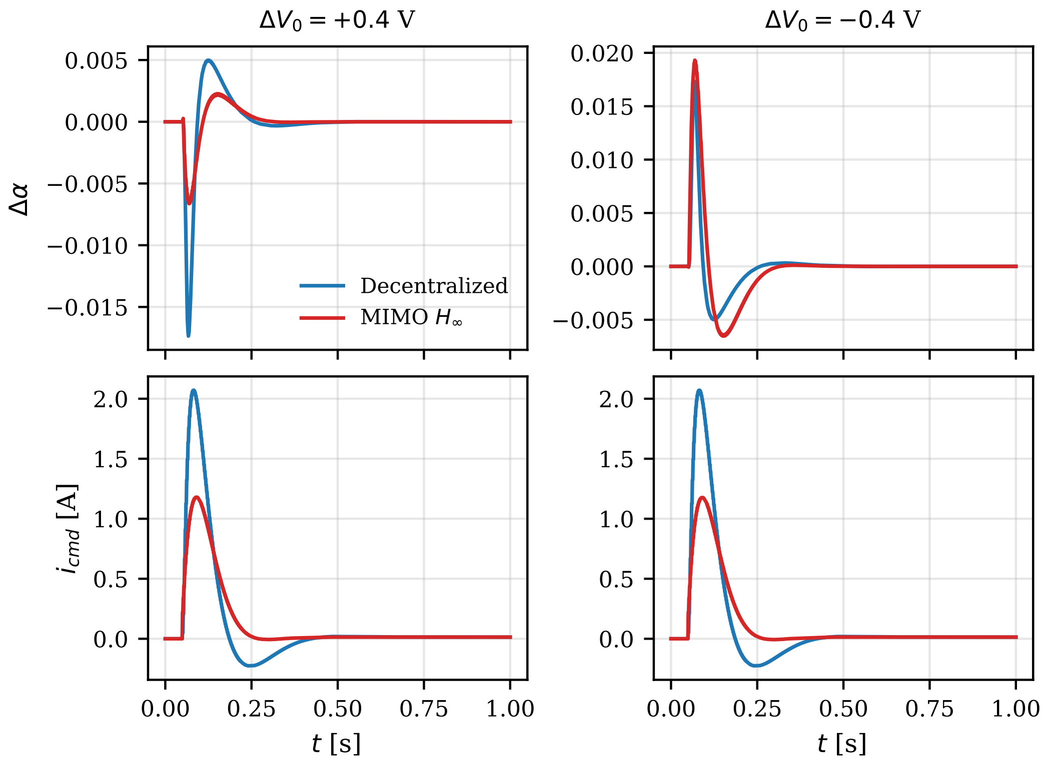

Figure 3. Small-signal drive transient, $\Delta v_{ref} = +0.05\,\mathrm{m\,s^{-1}}$ (zero time on the current rail). Left column $\Delta V_0 = +0.4\,\mathrm{V}$, right column $\Delta V_0 = -0.4\,\mathrm{V}$; top row share deviation, bottom row motor current command. The first 1 s of the 12 s simulation is shown; both responses are in steady state for the remainder. The decentralized traces are identical between columns while the centralized traces invert.

6.2.3. Metric 3: regen bus excursion

Run against the 15-state truth model with i_{cmd} railing at -20 A, the bus excursion is 1.4911 V against 1.4743 V, a ratio of 0.989: **essentially controller-independent**. The event is *not* rail-dominated — the current command spends only 0.18 to 0.34 % of the horizon on the clamp — so "both controllers sit on the same rail" does not account for the tie. What accounts for it is that the bus excursion is set by the *peak braking current the loop reaches*, and both controllers reach essentially the same peak on the same $-2\,\mathrm{m\,s^{-1}}$ demand; the droop loop, whatever its architecture, has no authority over a disturbance injected downstream of it on that timescale. This metric was proposed as an architecture discriminator [9]; **it is not one** — it measures the excitation, not the controller. On the same event the centralized controller is the better of the two on share excursion (0.1448 against 0.1747) and recovers the bus faster (0.43 s against 0.58 s) (Appendix D).

6.2.4. Input sensitivity $\bar{\sigma}(S_u)$

For a MIMO plant, output sensitivity does not bound input sensitivity: $S_u = (I + G_c G_p)^{-1}$ and $S_y = (I + G_p G_c)^{-1}$ differ whenever the plant condition number differs from unity, and the available bound ([3] §12.7.2, eqs. (12.86)–(12.87), p. 416) only gives $\bar{\sigma}(S_u) \le K(G_p)\,\bar{\sigma}(S_y)$. That is a live concern here, since $\mathrm{cond}(G_s)$ is 21.31 at DC and peaks near 8835 in band (Table 2), so the loose bound admits input sensitivities three orders of magnitude above the output ones. Direct evaluation (same continuous controller assembly as every other frequency metric in this report) shows the bound is far from tight:

Table 7. Input sensitivity $\bar{\sigma}(S_u)$, peak value and the frequency at which it occurs. Worst corner is $\Delta V_0 = -0.4\,\mathrm{V}$, $T_d = 2\,\mathrm{ms}$, $\tau_r = 300\,\mu\mathrm{s}$, $\tau_f = 0.8\,\mathrm{ms}$, $K_v = 2$, damping factor 2, $\tau_v = 0.5\,\mathrm{ms}$, $T_{d,v} = 4\,\mathrm{ms}$.

Case	`dec`	`mimo`
nominal, peak $\bar{\sigma}(S_u)$	1.3852 @ 35.5 rad s^{-1}	**1.2385** @ 33.1 rad s^{-1}
worst corner, peak $\bar{\sigma}(S_u)$	1.8640 @ 54.6 rad s^{-1}	**1.4749** @ 41.3 rad s^{-1}

Both controllers keep the realized input sensitivity modest — comparable to their output sensitivities rather than to the $K(G_p)$-scaled bound — and the centralized controller is again the better of the two, by 11 % nominally and 21 % at the worst corner. **The input node raises no new concern for either architecture**; the decentralized baseline's output-sensitivity deficit does not appear as an additional cost at the actuator.

6.2.5. Caveats

Six qualifications bound the above, expanded in Appendix D. (i) The ΔV_0 sign uncertainty bounds any feedforward benefit: the entire centralized coupling advantage lives in (7), zero at $\Delta V_0 = 0$ and sign-flipping with it. (ii) The K-out-of-envelope corners are waived; both controllers degrade badly there ($\bar{\sigma}(S_o) = 4.93$ centralized against 5.77 decentralized). (iii) The large steps and drive-cycle ramps still touch the clamp, so those comparisons are partly about anti-windup recovery rather than linear design. (iv) The Tier-2 worst-corner numbers are a representative subsample; the relative verdict survives, the absolute value does not. (v) In the FC-charge-cruise regime the droop ratio is parked on its upper clamp and neither controller shows windup interaction. (vi) **Nothing here is bench-calibrated**: ΔV_0 above all, plus k_t, the encoder chain, m_{eff}, τ_v and $T_{d,v}$ remain open, and a calibrated plant could move the verdicts, particularly the second metric.

7. Conclusions and Future Work

1. **The coupling is real, one-directional, and operating-point dominated.** The droop ratio does not move the wheel speed ($G_{21} \equiv 0$, a modelling result with its residual measured at 0.44 % of the bus), so the plant is exactly upper triangular and the RGA is identically I at every frequency and corner. The live path is drive to share, through $\partial\alpha/\partial I_{tot} = -\Delta V_0 r_0(1-r_0)/(k_d I_{tot0}^2)$, at 1.57× the direct path at 2 A and 25× it at 0.5 A on the normalized plant. **The RGA is the wrong instrument for this plant.**
2. **Decentralized control is justified, not merely convenient.** It is stable at all 4992 feasible Tier-1 corners, continuous and in the exact multirate discrete loop; it keeps a 2.3× whole-cycle speed-tracking advantage and a 45× faster recovery from saturation; it incurs a bounded penalty in worst-corner sensitivity (2.55 vs. 1.88 on the same corner set); and none of its advantages depend on an uncalibrated sign. The centralized controller's own gains say the same thing from the other side — it is about 98 % decentralized, with a 2.15 % coupling feedforward that the synthesis deliberately keeps small because it cannot trust the coupling's sign.
3. **The three primary metrics give three different answers**: the centralized controller is superior on worst-corner sensitivity by 27 %, ties on regen bus excursion (a metric that measures the excitation rather than the controller and should be retired), and is superior or inferior on the drive-transient share excursion depending on the sign of ΔV_0 — with a median penalty ratio $\approx 2.3\times$ between the opposing and matching signs. **The decentralized pair is not superior in every saturated large-signal case**: the large-signal share excursion splits by ΔV_0 sign exactly as the small-signal one does. What decentralized retains outright is drive-cycle tracking, post-saturation recovery speed, and sign-independence.

4. **The MIMO Youla-H $T(0) = I$ correction is effective and inexpensive to apply.** A constant right multiplier $M_{YH} = [G_s(0)Y_H(0)]^{-1}$ on the Youla parameter generalizes the scalar gain-tuning recipe without touching poles or zero dynamics, reduces to it exactly in the 1×1 case, and combined with an exact rank-2 integrator split makes $T(0) = I$ structural rather than numerical — it survives truncation, discretization and float32 quantization ($\|T(0) - I\| = 6.5 \times 10^{-10}$; float32 replay error 5.52×10^{-6}).
5. **Compute cost does not discriminate**: 44.5 against 49.5 kMAC/s, a 10 % margin nominally favouring the centralized controller only because the decentralized drive half cannot be a biquad cascade and needs a Hanus self-conditioned state-space form at a second rate. Both are under 0.1 % of the core. The genuine costs of centralization are halving the share loop rate, coupling the two control tasks' scheduling, and a matrix-valued anti-windup scheme.
6. **The actuator limit is a design node, not a datum.** It enters the input scaling D_u, so it sets how much physical current a given control-effort weight actually penalizes; choosing the motor clamp so that it binds together with the converter power budget ($\pm$20 A motor-side $\approx$ 4.9 A bus-side at cruise) makes every saturated result in this report a statement about the powertrain rather than about an arbitrary number, and the effort weights must be re-tuned whenever it moves.

7.1. Independent validation

The whole chain — plant assembly, H_∞ synthesis, the MIMO Youla-H $T(0) = I$ correction, the emitted coefficient header, the corner battery and the transients — was rebuilt independently in MATLAB against the shipped $\pm$20 A design and **passes all six acceptance criteria** (Appendix E). The design plant agrees to 2.23×10^{-9}, the a-posteriori $\|T_{zw}\|_\infty$ to 0.006 % (1.6457 against 1.6456), $T(0) = I$ holds exactly both in synthesis and on the parsed header, all 576 corners are stable, and the transient excursions agree to within 5 %. Two limitations reported above are consequences of that cross-check: the Tier-2 worst-corner figure is a representative-subsample lower bound — MATLAB's exhaustive sweep finds 1.9443 against the subsample's 1.8754, reproduced by the Python model at 1.9445 — and the cross-transfer improvement must be quoted on the implemented 2 ms loop (1.109×10^{-2}, matched exactly by both tools) and not in continuous time.

7.2. Future work

(1) Measure ΔV_0 — a direct reading of two no-load bus setpoints that collapses the sign uncertainty bounding the entire centralized advantage. Nothing else here is worth more per unit of bench time; a re-synthesis at the measured sign could then justify a materially larger feedforward. **(2) Close the drive-channel calibration**: tire radius and encoder chain first (inexpensive to measure, blocking for everything downstream), then k_t (requires selecting a motor), then τ_v and $T_{d,v}$ from a VESC current-step response, then the free-run current at 16 V which sets 93 % of b_{eff}. If calibration narrows the K_v envelope, re-run the drive weights with a lighter effort penalty and expect a materially faster drive channel. **(3) Address the** K**-envelope gap**: the SISO share controller was never gated at the light-load and FC-cruise points where K leaves $[0.55, 1.45]$, and both controllers degrade sharply there ($\bar{\sigma}(S_o) = 5.77$ decentralized, 4.93 centralized); gain scheduling on I_{tot} would help both architectures. **(4) Firmware integration is a separate exercise.** The emitted coefficient header is deliberately not wired into the build, and on the evidence here there is no operational case for wiring it in before ΔV_0 is known.

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Data Availability Statement: All numbers in this report are generated artefacts of the internal `controller_design_MIMO/` toolchain and are reproducible from the scripts and metric files referenced in [10–12].

Conflicts of Interest: The author declares no conflicts of interest.

Appendix A. Full-Order Truth-Model Validation

The design plant is checked against a 15-state converter-level truth model built by grafting the drive branch onto a verbatim copy (reproduced to 0.0 relative error) of the 11-state TPS61288 small-signal model used in the SISO work. That model is structurally matched — both channels share a reference and divider network — so ΔV_0 enters only through the realized current split: the droop conductances stay set by the commanded r_0 while the actual split becomes $\alpha_0 = r_0 + \Delta V_0 r_0 (1 - r_0)/(k_d I_{tot})$. That single change makes the coupling *emerge from the circuit*:

$$\delta\alpha = \frac{I_{B0}\,\delta i_F - I_{F0}\,\delta i_B}{I_{tot}^2}, \quad \delta i_F = -\frac{\delta v_{bus}}{k_d/r}, \quad \delta v_{bus} = -\delta I_{load}\,k_d \implies \frac{\delta\alpha}{\delta I_{load}} = -\frac{\Delta V_0\,r(1-r)}{k_d\,I_{tot}^2}, \tag{A1}$$

identical to (7). No coupling term is written into the truth model; comparing its DC $\partial\hat{\alpha}/\partial I_{load}$ against the design plant's analytic value gives agreement to **0.032 %** on the odd-in-ΔV_0 part, over $I_{tot} \in \{0.5, 2, 5\}$A $\times\, r_0 \in \{0.3, 0.5, 0.7\} \times \Delta V_0 \in \{0.2, 0.4\}$V $\times\, C_{bus} \in \{30, 500\}$µF.

A residual even-in-ΔV_0 offset of $\leq 2.4 \times 10^{-3}$ share per ampere is not a modelling error but a real asymmetry: the truth model's channels differ ($V_{in} = 9$ V vs. 8 V give different duty, compensation gain and right-half-plane zero) and the error amplifiers have finite DC gain, so a load step splits marginally unevenly even with matched references. Channel-by-channel frequency-response deviation over the design band is 0.93 % (share), 0.71 % (coupling) and 0.00 % (drive, shared construction) at nominal, worst 7.71 % over the full envelope against a 15 % tolerance. Design-plant against truth-model $\bar{\sigma}(S_o)$ at nominal agrees to 4.1 % (1.1606 vs. 1.2096) against a 20 % tolerance. One residual limitation: the truth-model $\|\cdot\|_\infty$ check falls back to a dense singular-value sweep, because the truth model mixes 0.12 rad s^{-1} vehicle dynamics with $\sim 10^5$ rad s^{-1} converter poles and the Hamiltonian bisection fails on that spread. A sweep is a lower bound — the conservative direction for this comparison — but not a proof.

Appendix B. Drive-Channel Weight Selection

Appendix B.1. Why the papers' second-order sensitivity weight fails on this plant

A second-order strictly-proper S weight with corner $a = \omega_c/\sqrt{dc - 1}$ demands $|S| \sim s^2$ roll-in below a, i.e. *two* integrators in the loop. The drive plant's mechanical pole sits at 0.122 rad s^{-1}, so below that frequency the plant contributes no integration and only the controller's single integrator is available; the weight requires an integral crossover around 1200 rad s^{-1} to serve a 24 rad s^{-1} loop, a 50× mismatch. Sweeping the W_p DC gain from 10^4 down to 4, the W_Δ break from 60 to 250 rad s^{-1} and the W_u DC from 0.3 to 0.04 never brought γ below about 1.2, with $\|W_p S\|$ binding at the low-frequency end. The papers' powertrain plant is a first-order lag whose pole is well above the weight corner, so the double-slope demand is served there; on SC001, with an 8 s mechanical time constant, it is not. This is a finding about the recipe's applicability domain; it is no defect in the recipe, and both the SISO and MIMO drive designs reach it independently.

Appendix B.2. SISO drive-channel iteration

Table A1. Drive-channel weight iteration (SISO baseline). W_Δ break fixed at 2.5× the W_p corner throughout; $W_u = \texttt{makeweight}(dc, 300, hf)$. The chosen row is the most aggressive setting that still clears the 45° phase-margin requirement — a deliberate relaxation of the 60° design guideline of [3] (p. 173), taken because this loop must hold across a wide corner family — and keeps worst-corner $\|S\|_\infty$ under the 2.5 target.

W_p corner [rad s^{-1}]	W_u	γ_{opt}	ω_c [rad s^{-1}]	PM [°]	worst $\|S\|_\infty$
24	$(0.3, 300, 20)$	9.14	12.3	56.6	1.64
24	$(0.1, 300, 5)$	5.60	15.2	46	—
40	$(0.3, 300, 20)$	13.25	16.9	54	1.89
50	$(\mathbf{0.2}, \mathbf{300}, \mathbf{10})$	**13.03**	**20.75**	**48.9**	**2.41**
60	$(0.1, 300, 5)$	—	24.4	42 (fails)	—

Appendix B.3. MIMO drive-block iteration

Table A2. Drive-block iteration for the centralized design, against the MIMO corner battery.

Iteration	Change	Outcome
it.1	papers' 2nd-order S weight	$\gamma = 1.2164$ — rejected, structurally unmeetable (see above)
it.2	1st-order family, same 24 rad s^{-1}	$\gamma = 0.9451$ — rejected, Tier-2 worst $\bar{\sigma}(S_o) = 2.17$
it.3a	relax W_u to $(0.15, 600, 20)$	$\gamma = 0.8349$ — *better* γ, rejected anyway (Tier-2 worst 2.41)
it.3b	W_u DC 0.3 → 0.5	heavier effort penalty retained; break frequency then swept below
it.4	W_u break → 18 rad s^{-1}	$\gamma = 1.6429$ — **final** (Table A3)

In iteration 3a a *lower* γ produced a *more aggressive* controller whose worst-corner sensitivity was worse. The final setting deliberately accepts a worse γ for corner margin, paying about 60 ms of small-signal drive settling. The reason a heavy control-effort penalty is the right instrument here is structural: the drive channel's uncertainty is $K_v \in \{0.5, 1, 2\}$ times a damping factor of $\{0.5, 2\}$, a $\approx 4\times$ gain spread, against the share channel's $K \in [0.55, 1.45]$; and the break frequency must sit strictly *below* the worst-case transport delay's phase crossover ($T_{d,v} \in [1, 4]$ms, i.e. 250 rad s^{-1} to 1000 rad s^{-1}).

Table A3. Selection of the drive W_u (DC gain, break frequency), swept against the Tier-2 in-envelope gate $\bar{\sigma}(S_o) < 2.0$ and Tier-1 stability. Only the break frequency was moved from its it.3b value; the chosen row clears the gate with about 6 % margin while keeping the break as close as possible to the 24 rad s^{-1} W_p corner — pushing it lower provides margin but places W_u and W_p in direct contention over the same decade.

W_u DC	break [rad s^{-1}]	γ_{MIMO}	Tier-1 unstable	Tier-2 worst $\bar{\sigma}(S_o)$
0.5	200	1.08	**61**	2.97
0.5	100	1.16	0	2.74
0.5	50	1.30	0	2.37
0.9	25	1.42	0	2.19
0.5	25	1.51	0	2.01
0.9	14	1.56	0	1.97
0.5	**18**	**1.65**	**0**	**1.88**
0.5	12	1.83	0	1.74

The corner result is sensitive to the balanced truncation tolerance, since a loose tolerance can leave truncation error that happens to damp the worst corner. Tightening the tolerance to 10^{-2} makes the reduced controller a faithful reduction and the corner result a property of the design rather than of the reduction — at the cost of two extra states (Section 5), which is the less expensive of the two options.

Appendix C. DGKF Implementation Details

The synthesis solves the standard two-Riccati H_∞ problem [8] — stated in the notation used here as Thm. 15.2, eqs. (15.94)–(15.98), pp. 491–493 of [3] — on the 14-state augmented plant with $p_z = 6$, $p_y = 2$, $m_u = 2$, using a Hamiltonian/Schur Riccati solver and a γ-bisection. The control and filter Riccati solutions are written P_c and P_f (not the more common X and Y), because Y is reserved throughout this report for the Youla transfer function. The Remark on p. 492 of the same source — a control input entering the plant without a weight violates the D_{12} full-column-rank condition and renders the problem unsolvable — fixes W_u as mandatory on both channels.

Appendix C.1. The P_f-Riccati degeneracy

With measurement-equals-reference structure $D_{21} = I_2$, so $Q_f = B_1(I - D_{21}^\top D_{21})B_1^\top = 0$ and $A_f = A - B_1 C_2$. In the augmented plant B_1 injects only into the W_p block row, so $A - B_1 C_2$ cancels that coupling exactly and is block-triangular with stable diagonal blocks: $P_f = 0$ is *provably* the stabilizing solution, and a correct two-Riccati implementation must return it. Ours does, to 1.0×10^{-14}, validating the estimation half of the code path on a problem that does not otherwise exercise it.

Appendix C.2. Optimism of the Riccati γ in MIMO

Because $P_f \equiv 0$, the DGKF feasibility test $\rho(P_c P_f) < \gamma^2$ is vacuous, leaving only "P_c-Riccati solvable and positive semi-definite" — necessary but here not sufficient. The bisection reports $\gamma = 0.89769$; the central controller attains $\gamma_{achieved} = 1.64563$, with $\|T_{zw}\|_\infty = 1.64573 \leq 1.005\gamma$. In the 1×1 sub-problems the bisection is exact to four decimals, so the gap is a MIMO-only artefact of the degeneracy. A refined back-off bisection on the a-posteriori criterion is therefore used so the delivered level is tight, and every γ quoted in this report is an a-posteriori achieved level. The same pattern was confirmed independently in MATLAB, whose `hinfsyn` result landed between the bisection value and the a-posteriori level exactly as the degeneracy predicts (Appendix E).

Appendix C.3. Scaling

All synthesis is performed on $G_s = D_e^{-1} G D_u$ with $D_e = \mathrm{diag}(0.05, 0.5)$ (the maximum acceptable share and speed errors) and $D_u = \mathrm{diag}(0.35, 20.0)$ (half the r span and the motor-current clamp), following [2]; the supporting rationale is the renormalization-of-gains argument of [3] §4.4.1, p. 159 — a gain is only meaningful relative to the actuator limit it must respect, which is exactly what D_u encodes. The scalings are kept *outside* the emitted controller matrices, both so the anti-windup back-calculation map is exactly D_u^{-1} and so float32 coefficient rounding is not applied to numbers spanning two decades of physical units. D_u carries the actuator limit into the cost function, so **the control-effort weight W_u penalizes a *normalized* input, and the physical current a given W_u actually penalizes scales with the clamp**. Changing the actuator limit therefore changes the effective effort penalty and requires re-tuning W_u — it is not a post-processing parameter.

Appendix D. Corner Families and Full Comparison Results

Appendix D.1. Corner families and feasibility

The corner families are 10 operating points ($\{0.5, 2, 5\}$A $\times$ $\{0.3, 0.5, 0.7\}$ plus the FC-charge-cruise point ($2\,\mathrm{A}, 0.85$)), 24 share corners ($\Delta V_0 \in \{-0.4, 0, +0.4\}$V, $T_d \in \{0.5, 2\}$ms, $\tau_r \in \{20, 300\}$µs, $\tau_f \in \{0, 0.8\}$ms), and 24 drive corners ($K_v \in \{0.5, 1, 2\}$, damping factor $\in \{0.5, 2\}$, $\tau_v \in \{0.5, 5\}$ms, $T_{d,v} \in \{1, 4\}$ms). ΔV_0 lives on the uncertainty axis rather than the operating axis and overrides the operating point's value when the two are combined.

The full Tier-1 cross product is $10 \times 24 \times 24 = 5760$ corners, of which **768 are infeasible and are skipped, counted, and never silently linearized**: the ideal-diode switches are unidirectional, so if (2) puts α_0 outside $(0, 1)$ the weaker source is blocked and the linearization is meaningless there. At 0.5 A the mismatch term reaches ± 0.56 share, so $(0.5\,\mathrm{A}, 0.5, \pm 0.4)$ gives $\alpha_0 = 1.17$ or -0.17. The remaining **4992 feasible corners** are the Tier-1 family used throughout. Inside the SISO share domain ($I_{tot} \geq 2\,\mathrm{A}$, $r \in [0.3, 0.7]$) the implied K envelope is $[0.733, 1.267]$, comfortably inside the SISO design's $[0.55, 1.45]$; over the full feasible operating set it is $[0.533, 2.067]$, reaching 2.067 at light load with full mismatch ($0.5\,\mathrm{A}$, $r_0 = 0.3$, $\Delta V_0 = +0.4\,\mathrm{V}$), 43 % above the SISO envelope. Those corners are carried here as stability-only corners.

Appendix D.2. Tier-1 stability

Table A4. Tier-1 stability, 4992 feasible corners. The discrete check for `dec` is not an approximation: it is the exact two-step monodromy of the 1 ms-lifted periodic loop.

Controller	cont. unstable	worst $\mathrm{Re}(\lambda)$ [rad s^{-1}]	disc. unstable	worst $\lvert z \rvert$
`dec` (multirate)	**0 / 4992**	-0.00400	**0 / 4992**	0.999992
`dec500`	—	—	**0 / 4992**	0.999992
`mimo` (single-rate)	**0 / 4992**	-0.11999	**0 / 4992**	0.999760

The worst discrete $|z| = 0.999760$ for `mimo` is the exact integrator at the sample rate, not a marginal design pole. The decentralized pair's slowest closed-loop mode is $0.0040\,\mathrm{rad\,s^{-1}}$ (a 250 s time constant) against the centralized loop's $0.1198\,\mathrm{rad\,s^{-1}}$, a factor of 30: both are stable, but the decentralized pair leaves a very slow, nearly-cancelled mode that the centralized design does not.

Appendix D.3. Tier-2 sensitivity

Table A5. Sensitivity at the nominal point and over the Tier-2 family (168 in-envelope corners of 208, the remainder in two documented waiver families). The waiver policy is identical for both controllers so they are judged on the same terms.

Quantity	`dec`	`mimo`	**verdict**
Nominal plant:			
$\|S_o\|_\infty$ ($\bar{\sigma}$ peak)	1.3974	**1.2287**	MIMO
$\|S_{11}\|$ peak (share)	1.2523	**1.2114**	MIMO
$\|S_{22}\|$ peak (drive)	1.3851	**1.2285**	MIMO
$\|T_{\alpha\leftarrow v_{ref}}\|_\infty$ [share/(m/s)], continuous	0.25676	**0.00131**	MIMO (195×)
same, on the implemented 2 ms loop	0.25676	**0.01109**	MIMO ($\sim 23\times$)
slowest closed-loop mode [rad s^{-1}]	0.00400	**0.1219**	MIMO
Tier-2, 168 in-envelope corners:			
worst $\bar{\sigma}(S_o)$	2.5535	**1.8754**	MIMO, $-27\,\%$
worst $\|S_{11}\|$ peak	1.5769	**1.3721**	MIMO
worst $\|S_{22}\|$ peak	2.4062	**1.6743**	MIMO, $-30\,\%$
worst $\|T_{\alpha\leftarrow v_{ref}}\|_\infty$	0.8480	**0.8020**	MIMO
continuous-unstable corners	0	0	tie
corners where MIMO has lower $\bar{\sigma}(S_o)$	**167 of 168**		MIMO
Waived families (stability-gated only):			
worst $\bar{\sigma}(S_o)$, FC cruise ($r_0 = 0.85$)	2.4960	**1.7471**	MIMO
worst $\bar{\sigma}(S_o)$, K-out-of-envelope	5.7702	**4.9334**	MIMO

The Tier-2 worst $\bar{\sigma}(S_o) = 1.8754$ is computed on a representative corner set and is a lower bound on the full-grid worst: an exhaustive 24×24 sweep at the nominal operating point locates worse corners outside the representative set (Appendix E reports one such sweep). The consequence is that a "< 2.0" absolute performance target read off this table is subsample-dependent and should not be quoted as a certificate. The decentralized comparison point (2.5535) comes from the same representative set, so the *relative* 27 % verdict stands, and stability is not at issue either way (Tier-1 is 0/4992 for both).

The input-node counterpart (Table 7) is computed on the same continuous controller assembly and at the same worst corner: peak $\bar{\sigma}(S_u)$ is 1.3852 (dec) against 1.2385 (MIMO) at the nominal plant and 1.8640 against 1.4749 at the worst corner. The input sensitivities therefore track the output ones closely rather than approaching the $K(G_p)\,\bar{\sigma}(S_y)$ bound of [3] §12.7.2, and the relative verdict is unchanged at the input node.

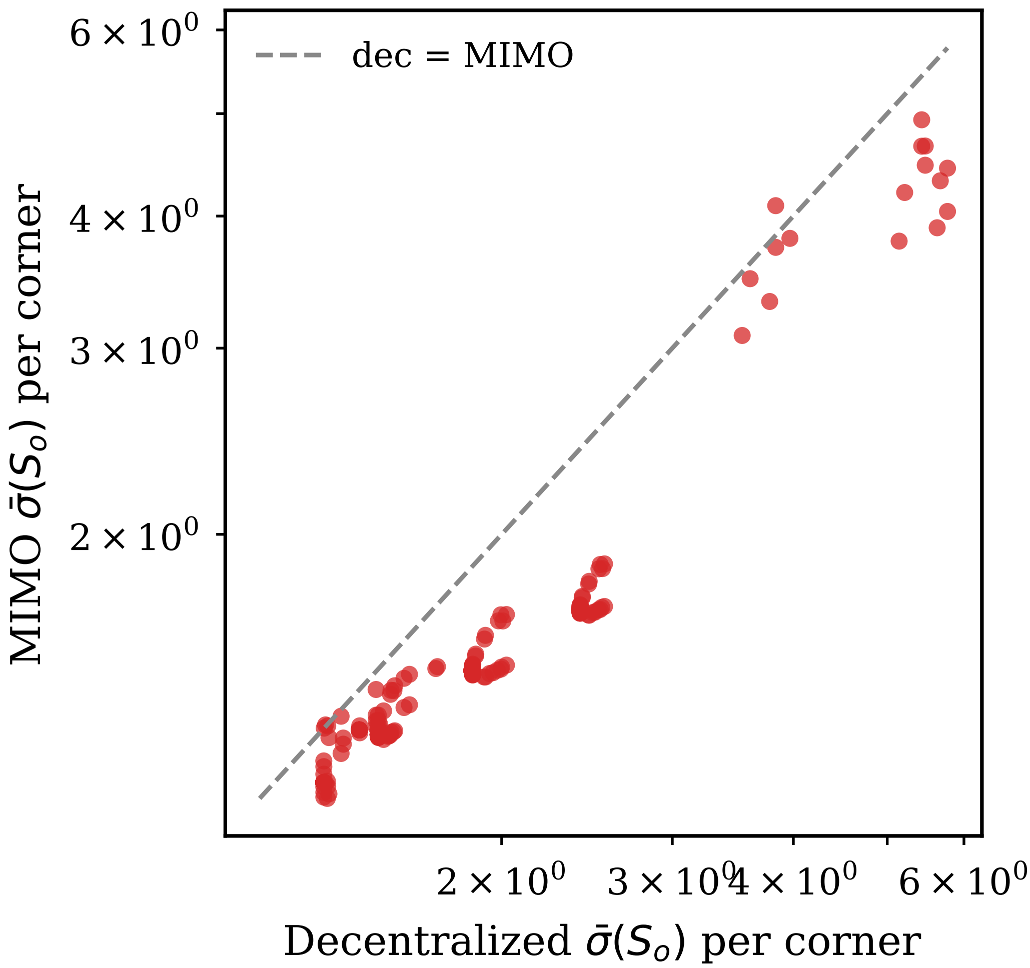

Figure A1. Per-corner $\bar{\sigma}(S_o)$, decentralized (horizontal) against MIMO (vertical), with the $y = x$ reference line. Points below the line favour the centralized controller: 167 of 168 in-envelope corners lie there. The out-of-envelope corners — light load with full mismatch, where the share plant gain reaches $K \approx 2.07$ — are the visible cluster in which both controllers degrade, the decentralized one slightly more.

Appendix D.4. Drive-transient share excursion, in full

A speed-reference step is applied at $t = 0.05\,\mathrm{s}$. The small step ($\Delta v_{ref} = +0.05\,\mathrm{m\,s^{-1}}$) is fully linear — zero time on the current rail — and is the clean comparison. The large step ($+2\,\mathrm{m\,s^{-1}}$) puts the loop on the $\pm 20\,\mathrm{A}$ rail for 0.18 to 0.34 % of the horizon: brief, but enough to make the comparison partly an anti-windup-recovery comparison and no longer a purely linear one. The 60 s horizon was chosen so the slow centralized recovery is measured outright and not reported as a horizon artefact.

Table A6. Drive-transient share excursion. Amplitudes are deviations about $v_0 = 2\,\mathrm{m\,s^{-1}}$; horizons are 12 s (small) and 60 s (large).

Quantity	`dec`	`dec500`	`mimo`
Small signal, $\Delta v_{ref} = +0.05\,\mathrm{m\,s^{-1}}$ (0 % rail):			
max $\lvert\Delta\alpha\rvert$, $\Delta V_0 = +0.4$	0.01744	0.01868	**0.00667**
max $\lvert\Delta\alpha\rvert$, $\Delta V_0 = -0.4$	**0.01744**	0.01868	0.01935
$\Delta\alpha$ settle (< 0.002) [s]	0.189	0.189	**0.175** (+) / 0.235 (−)
v settle (2 %) [s]	0.358	0.358	**0.199**
peak $\lvert i_{cmd}\rvert$ [A]	2.071	2.071	**1.179**
MIMO/dec excursion ratio	—	—	**0.383** (+) / **1.109** (−)
Large signal, $\Delta v_{ref} = +2\,\mathrm{m\,s^{-1}}$ (touches the clamp):			
max $\lvert\Delta\alpha\rvert$, $\Delta V_0 = +0.4$	0.3306	0.3334	**0.1729**
max $\lvert\Delta\alpha\rvert$, $\Delta V_0 = -0.4$	**0.3306**	0.3334	0.6289
$\Delta\alpha$ settle (< 0.002) [s]	0.599	0.601	**0.337** (+) / 0.461 (−)
v settle (2 %) [s]	**0.501**	0.501	22.45
droop span used, $\Delta V_0 = +0.4$	0.431–0.850	0.431–0.850	0.500–0.850
droop span used, $\Delta V_0 = -0.4$	0.150–0.569	0.150–0.569	**0.386–0.719**
rail fraction	0.00337	0.00337	**0.00187**

Appendix D.5. Regen event on the truth model

The regen event is run against the 15-state truth model with a bus-voltage output in place of the 8-state design plant. Δv_{ref} is stepped to $-2\,\mathrm{m\,s^{-1}}$ ($2\,\mathrm{m\,s^{-1}}$ to standstill) and i_{cmd} touches the -20 A rail for 0.18–0.34 % of the horizon; the 0.8 ms prefilter is applied digitally at the base rate since the truth model carries no τ_f by convention. Horizon 60 s.

Table A7. Regen event, 15-state truth model. The residual Δv_{bus} of 0.105 V is the new steady-state load operating point (the vehicle has stopped) and is no tracking error; recovery is therefore measured about the final value.

Quantity	`dec`	`dec500`	`mimo`
max $\lvert\Delta v_{bus}\rvert$ [V]	1.49112	1.49112	**1.47429**
Δv_{bus} final [V]	0.10478	0.10478	0.10477
v_{bus} recovery [s]	0.582	0.582	**0.431**
max $\lvert\Delta\alpha\rvert$	0.17467	0.17263	**0.14478**
peak i_{cmd} [A]	−20.00	−20.00	−20.00
rail fraction	0.00337	0.00337	**0.00183**
v settle (2 %) [s]	**0.501**	0.501	22.45

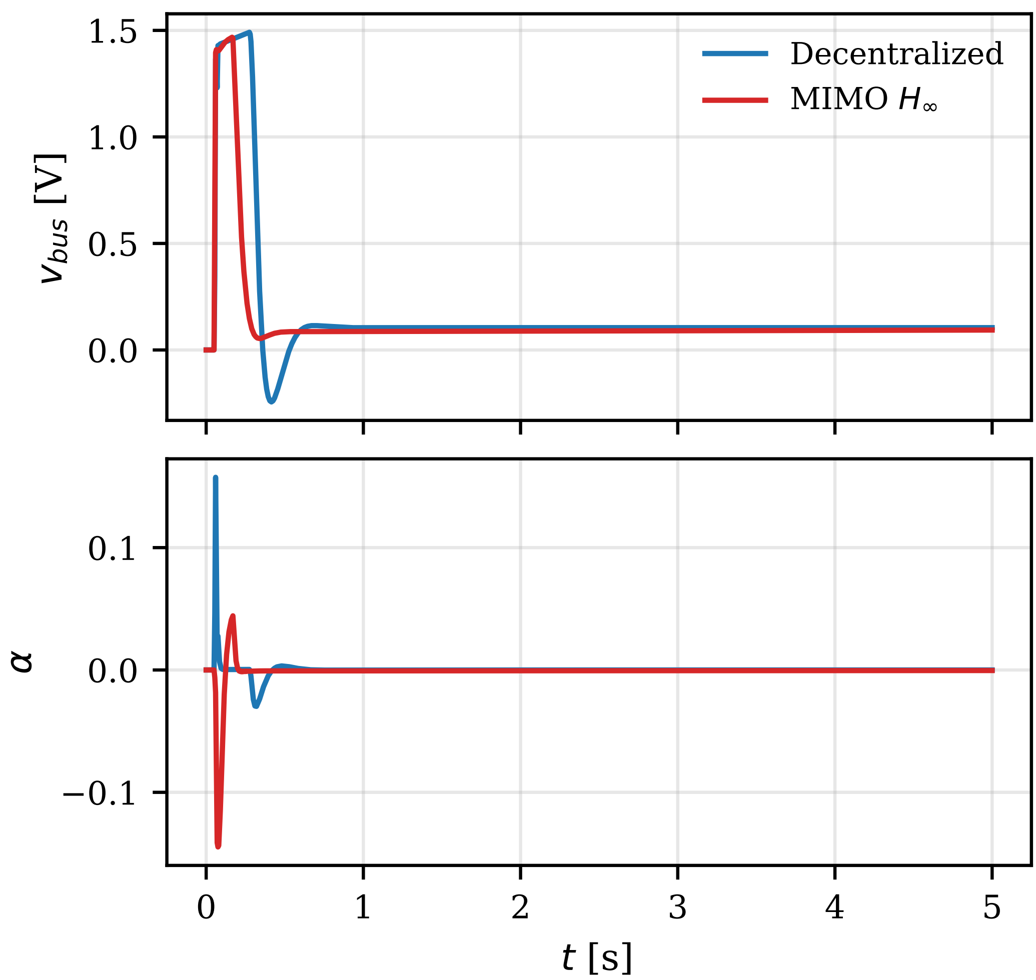

Figure A2. Regen event on the 15-state truth model; the first 5 s of the 60 s simulation is shown (the response is in steady state for the remainder). Top: bus-voltage deviation. Bottom: share deviation. The bus traces are nearly indistinguishable, and not because the event is rail-dominated (the clamp is touched for well under 1 % of the evaluation horizon): the excursion is set by the peak braking current both controllers reach, which is a property of the demand and of the plant alone.

Appendix D.6. FC-charge cruise and the drive cycle

At the FC-charge-cruise operating point (2 A, $r_0 = 0.85$, static share $\alpha_0 = 0.8925$) the energy-management strategy parks the droop ratio on its upper clamp so the fuel cell carries the bus and the battery charges; the share loop has no upward authority left by construction. A $+0.5\,\mathrm{m\,s^{-1}}$ drive step over a 60 s horizon shows **no windup interaction for either controller**: both use only the small downward authority available ($r_{min} = 0.83693$ and 0.84776), both converge to a $\sim 2 \times 10^{-3}$ residual share error — a direct consequence of the authority loss, with no sign of a hang — and neither shows a limit cycle (tail peak-to-peak $\leq 10^{-10}$). All three anti-windup schemes behave gracefully in the degenerate-authority regime. The two do differ in how long they spend pinned: the decentralized pair holds the upper clamp for 99.6 % of the horizon against 0.4 % for the centralized controller, and the centralized peak share excursion is correspondingly smaller (0.061 29 against 0.103 10, −41 %). Since the regime is defined by *having no upward authority*, that difference is a statement about how quickly each controller gives up on demanding it, and says nothing about disturbance rejection.

The 30 s drive-cycle profile (standstill; ramp 0 to $2\,\mathrm{m\,s^{-1}}$ over 5 s; cruise; coast to $0.5\,\mathrm{m\,s^{-1}}$; hold; stop) with a $\Delta\alpha_{ref} = +0.2$ share step over 11–15 s and a −1.0 A load disturbance over 22–25 s is where the decentralized advantage is largest: the whole-cycle

RMS speed error is 0.1059 $\mathrm{m\,s^{-1}}$ against 0.2396 $\mathrm{m\,s^{-1}}$, a factor of 2.3, and the RMS share error 0.004 75 against 0.007 30. The *cruise window* shows why: the decentralized pair is at numerical zero there (2.94×10^{-7} $\mathrm{m\,s^{-1}}$ RMS speed error) while the centralized controller is at 0.195 $\mathrm{m\,s^{-1}}$, still recovering from the ramp. On the isolated load disturbance the decentralized pair is also better by 2.2$\times$ in speed (0.0191 against 0.0420 $\mathrm{m\,s^{-1}}$) though nearly equal in share (0.0091 against 0.0097). The centralized controller drives the droop ratio to its *lower* clamp ($r = 0.15$) during the cycle, whereas the decentralized pair never goes below 0.29 — the cross-channel term spending droop authority in response to a saturated drive channel, a real cost of centralization on this plant.

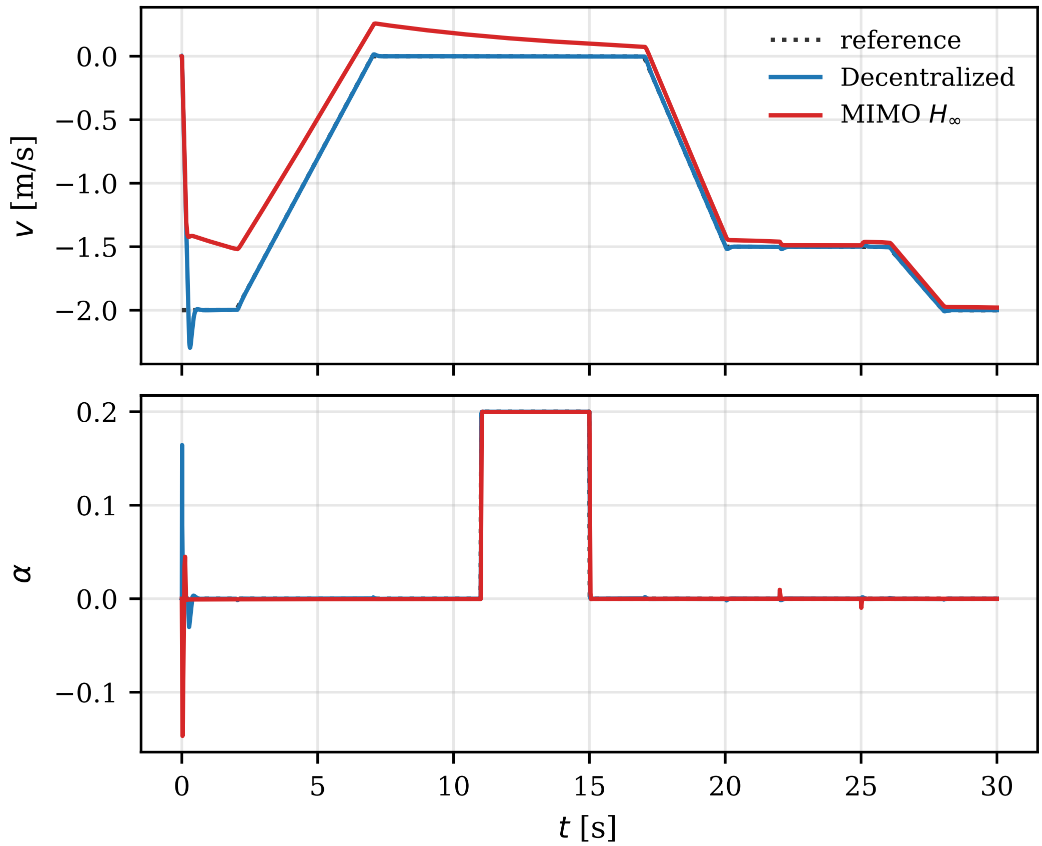

Figure A3. 30 s drive cycle. Top: speed against reference. Bottom: share against reference. The cruise segment (8–11 s) is where the gap is widest: the decentralized pair has settled while the centralized one is still recovering from the ramp.

Appendix D.7. Caveats in full

1. **ΔV_0 sign uncertainty bounds any feedforward benefit.** The entire centralized coupling advantage lives in (7), which is zero at $\Delta V_0 = 0$ and flips sign with it. Until ΔV_0 is measured, the defensible claim is that the centralized design improves coupling rejection at the design sign and degrades it at the mirrored sign.
2. **The K-out-of-envelope corners are waived, and both controllers degrade badly there** — $\bar{\sigma}(S_o) = 5.77$ decentralized against 4.93 centralized. The stability-only waiver is physically justified: these are the 16 light-load, full-mismatch corners where $K \approx 2.1$ and the static share sits within ~ 0.02 of the unidirectional-switch feasibility boundary, so the plant model itself is at the edge of validity. Neither absolute figure should be read as a performance claim.
3. **Large steps touch the clamp.** The 2 $\mathrm{m\,s^{-1}}$ step and the drive-cycle ramps reach the $\pm$20 A clamp for a fraction of a percent of the horizon; those comparisons are therefore partly about anti-windup recovery rather than about linear design alone.

4. **The Tier-2 worst-corner numbers come from a representative corner set**: the relative verdict survives, the absolute value does not.
5. **Nothing here is bench-calibrated.** Every calibration item remains open — ΔV_0 above all, plus k_t, the encoder chain, m_{eff}, τ_v and $T_{d,v}$. The corner families are wide precisely because of this, but a calibrated plant could move the verdicts, particularly the second primary metric.

Appendix E. MATLAB Cross-Validation

The entire chain was cross-checked in MATLAB (Control System and Robust Control Toolboxes) by a script that independently rebuilds the plant from the documented equations, re-runs `hinfsyn` and the MIMO Youla-H correction, parses the emitted coefficient header, and re-runs the corner battery and the transients. **The verdict is PASS on all six criteria** (Table A8).

Appendix E.1. Scope of the run

The cross-check was run against the shipped design, at the ± 20 A actuator span ($D_u = \text{diag}(0.35, 20.0)$) used throughout this report, so Table A8 validates the report as it stands — plant, synthesis, $T(0) = I$ correction, emitted coefficient header, corner battery and transients alike. An earlier cross-check at a ± 5 A span validated the same machinery and algebra at the previous actuator limit and first identified the two methodological findings recorded below (the representative-subsample bound on the Tier-2 worst corner, and the distinction between the continuous and the implemented cross-transfer); the ± 20 A run reconfirms both. Its numbers are superseded by those in Table A8 and are retained only in the design record [12].

Table A8. MATLAB cross-validation anchors for the shipped ± 20 A design. The right-hand column is the corresponding Python result. The two tools solve the same problem by independent implementations, so exact agreement is expected only where the quantity is algebraic; where it is recomputed (the γ bisection, a ZOH-discretized sensitivity peak) the acceptance criterion is stated with the row.

Anchor	MATLAB	Python
design-plant DC gain (max abs. deviation)	2.23×10^{-9}	
`hinfsyn` γ	1.0851	0.8977 (bisection) / 1.6456 (a-posteriori)
$\lVert M_{YH} - I \rVert_2$	9.631×10^{-5}	1.75×10^{-4}
$\text{cond}(G_s(0) Y_H(0))$	1.0000	1.0001
$T(0)$, in synthesis and on the parsed header	I exactly	
a-posteriori $\lVert T_{zw} \rVert_\infty$ of the emitted artefact	1.6457	1.6456 (0.006 %)
nominal $\bar{\sigma}(S_o)$	1.2623	1.2287 (2.7 %, criterion 5 %)
$\lVert T_{\alpha \leftarrow v_{ref}} \rVert_\infty$, implemented 2 ms loop	1.109×10^{-2}	1.109×10^{-2}
corner battery	576 / 576 stable	
exhaustive-sweep worst $\bar{\sigma}(S_o)$	1.9443	1.9445
share excursion, $\Delta V_0 = +0.4 / -0.4$	0.00636 / 0.01900	0.00667 / 0.01935
peak $\lvert i_{cmd} \rvert$ [A], $\Delta V_0 = +0.4 / -0.4$	1.179 / 1.176	1.179 / 1.176

The `hinfsyn` level of 1.0851 lands between the Python Riccati bisection (0.8977) and the a-posteriori achieved level (1.6456), which is precisely the behaviour the P_f degeneracy of Appendix C predicts: the bisection is optimistic because its feasibility test is vacuous, and a third implementation therefore has no reason to reproduce either endpoint. The two values of $\|M_{YH} - I\|_2$ (9.6×10^{-5} and 1.75×10^{-4}) are computed on separately synthesized controllers and are quoted as computed; both say the same thing, that the H_∞ controller was already integral to four decimals. And the nominal $\bar{\sigma}(S_o)$ differs by 2.7 % because MATLAB evaluates the ZOH-discretized loop where the Python figure is continuous; the criterion for that row is 5 %.

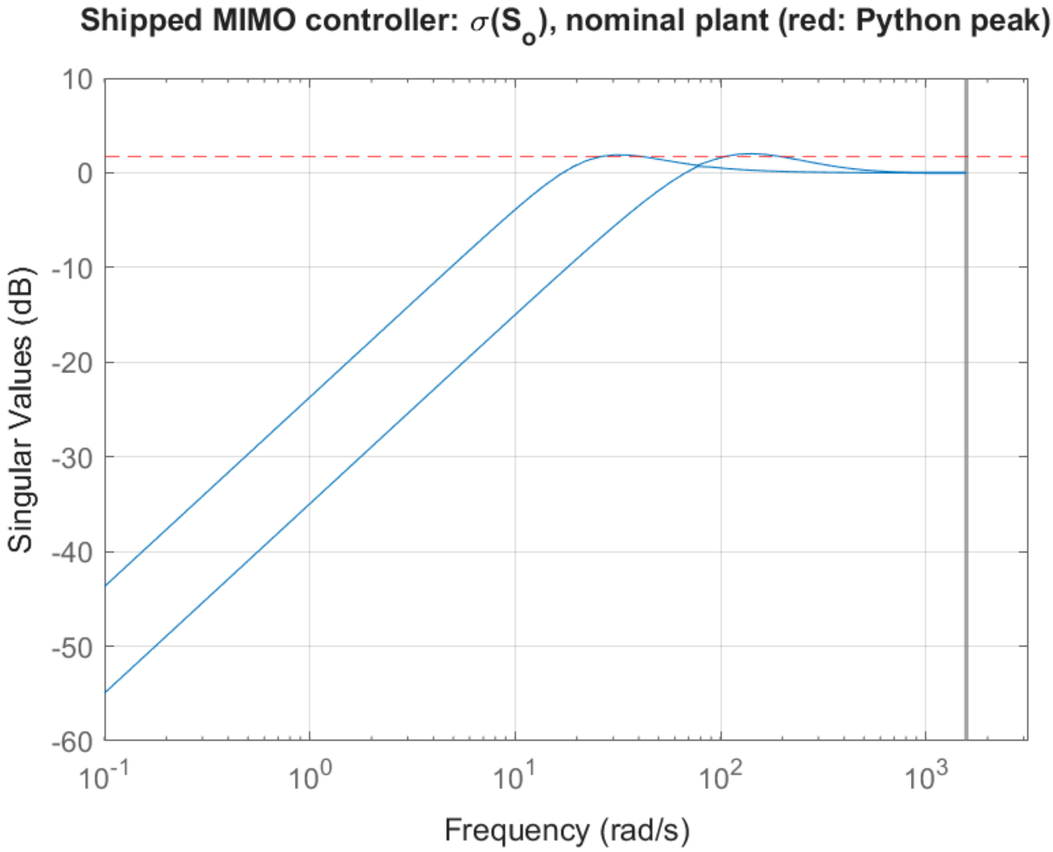

Figure A4. MATLAB cross-check of $\bar{\sigma}(S_o)$ versus frequency, built independently from the documented plant equations and the parsed coefficient header, for the shipped ±20 A design. Compare Figure 2: the shapes and the peak levels agree, the residual difference being that this curve is evaluated on the ZOH-discretized loop.

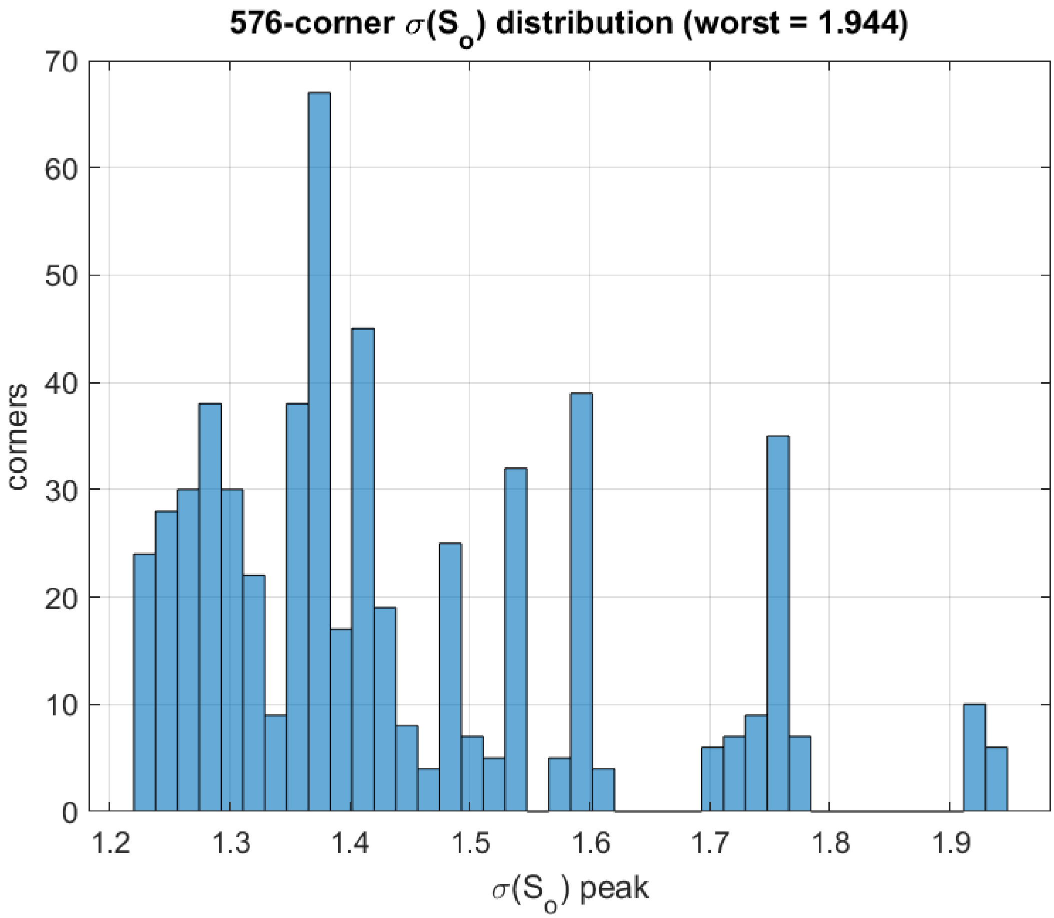

Figure A5. MATLAB cross-check of the per-corner robustness scatter (compare Figure A1), for the shipped $\pm 20\,\mathrm{A}$ design. This sweep is the *full* 24×24 family at the nominal operating point, where Figure A1 is the representative subsample; it is what establishes that the subsample misses worse corners.

Two methodological findings were first surfaced by the $\pm 5\,\mathrm{A}$ run and are reconfirmed here, so both are structural and no artefact of a particular actuator span. First, the MATLAB sweep is exhaustive at the nominal operating point, and it locates the worst $\bar{\sigma}(S_o)$ corner *outside* the Python representative corner set: $\bar{\sigma}(S_o) = 1.9443$ at $\Delta V_0 = -0.4$, $T_d = 2\,\mathrm{ms}$, $\tau_r = 300\,\mu\mathrm{s}$, $\tau_f = 0.8\,\mathrm{ms}$, $K_v = 2$, damping factor 2, $\tau_v = 5\,\mathrm{ms}$, $T_{d,v} = 4\,\mathrm{ms}$, against the representative-set figure of 1.8754 reported in Table A5. The Python model reproduces that corner independently at 1.9445, and stability is unaffected (576/576 here, 0/4992 on Tier-1). That is why the Tier-2 worst-corner figure is reported throughout as a subsample lower bound rather than as a certificate. Second, the nominal cross-transfer improvement is a *continuous-domain* figure and must be requoted on the implemented 2 ms loop; MATLAB obtains 1.109×10^{-2} there, matching the implemented-loop row of Table A5 exactly, which is why that table carries both rows. The two implementations agree everywhere they should agree, and the residual differences are attributable to the stated approximations alone.

Appendix E.2. Reproduction tolerance

The share-channel controller is re-derived here from its documented weights instead of parsed from the firmware header, and the re-derivation carries a 0.67 % integrator-gain offset (112.679 against 111.930) attributable to solver accuracy at the γ_{opt} anchor. The offset is well inside the corner family and affects no comparison in this report.

Appendix F. Anti-Windup and Discretization

For the standard configuration $\hat{u}/\hat{n} = -Y$ ([3] §3.8.2, eq. (3.124), p. 132): the Youla transfer function *is* the noise-to-actuator map. The W_u weight acting on Y in Table 3 is therefore a direct bound on actuator activity, and the anti-windup schemes below handle the residual demand that the weight does not remove.

Appendix F.1. Decentralized baseline: Hanus self-conditioning

The SISO drive controller's *non-integral* branch alone has a DC gain of $367.7\,\mathrm{A\,m^{-1}\,s^{-1}}$, so it saturates the $\pm 20\,\mathrm{A}$ actuator for speed errors above $e_{sat} = 54.4\,\mathrm{mm\,s^{-1}}$ and its lag states wind up independently of the integrator. The threshold scales with the clamp, so the scheme's adequacy is an amplitude question: integrator-only back-calculation is clean on steps up to $0.2\,\mathrm{m\,s^{-1}}$ (final errors of order $10^{-8}\,\mathrm{m\,s^{-1}}$, no limit cycle) and fails from $0.5\,\mathrm{m\,s^{-1}}$, producing a $-0.101\,\mathrm{m\,s^{-1}}$ standing error and a $6.3\,\mathrm{mm\,s^{-1}}$ limit cycle on the 0 to $2\,\mathrm{m\,s^{-1}}$ step. Because that step is a required gate, the baseline uses Hanus self-conditioning [7] on a five-state state-space realization, $x_{k+1} = (A_d - B_d C_d / D_d) x_k + B_d u / D_d$ with u the clamped output — exactly equivalent to the biquad-plus-integrator form when unsaturated (verified to $6.32 \times 10^{-10}\,\mathrm{A}$), but *not* implementable as a biquad cascade. This is the origin of the six implemented states in Table 5.

Appendix F.2. Centralized controller: reduction, discretization, modal form

The controller $G_{c,YH}$ has order 30 (14 controller + 8 plant + 8 from the Youla interconnection); after the integrator split the stable remainder is order 28. Balanced truncation selects order 7 at a 10^{-2} margin, with a maximum relative $\bar{\sigma}$ deviation of 1.88×10^{-3} over $\omega \in [0.1, 10^4]$ against a 2×10^{-2} tolerance. Deviation is measured per frequency instead of against a single global scale, which would be dominated by the integrator's low-frequency magnitude and would make the check vacuous. The final continuous controller is 9 states: two exact integrators plus a seven-state remainder.

The integrator is an exact 2×2 Tustin block with the matrix K_I, kept separate from the remainder so anti-windup can act on it; the recursive trapezoid form is verified equal to the LTI form at 2.8×10^{-17}. *Ordering matters*: the integrator is advanced first and the new value used in the output, matching the firmware idiom — advancing it afterwards inserts a hidden extra sample of delay, which the equivalence check catches. The stable remainder is discretized by Tustin and transformed to real modal (block-diagonal) form, $A_{modal} = \mathrm{diag}\left(-0.9866,\ 0.9225,\ 0.6205,\ \left[\begin{smallmatrix} -0.8592 & 0.0737 \\ -0.0737 & -0.8592 \end{smallmatrix}\right],\ \left[\begin{smallmatrix} -0.0995 & 0.2398 \\ -0.2398 & -0.0995 \end{smallmatrix}\right]\right)$, at $\mathrm{cond}(T) = 7.24$ and reproducing the Tustin response to 7.9×10^{-16}. The rationale is float32: a modal A has the pole locations themselves as its coefficients, so rounding perturbs each mode independently instead of perturbing companion-form polynomial coefficients, where a 10^{-7} relative error can move clustered roots by far more.

Appendix F.3. Centralized anti-windup: back-calculation plus authority clamp

Anti-windup has two parts. Because the integrator state adds directly to the scaled input, its output map is I_2 and the back-calculation map is exactly D_u^{-1}: on saturation, $x_{int} \mathrel{+}= D_u^{-1}\delta$ with $\delta = u_{sat} - u$ physical, and $\mathrm{cond}(D_u^{-1}) = 57.14$ so no diagonal fallback is needed — the directionality-preserving generalization of the scalar scheme. On top of it, x_{int} is clamped componentwise to $[-1, 1]^2$: the integrator alone may never demand more than the actuator can deliver.

The second term guards against bare back-calculation dumping the *whole* clamp excess into the integrator, including the part created by the direct feedthrough — which lets the integrator absorb a direct-term excess and then hold the *opposite* rail once the error collapses. The scalar share controller's direct term is small enough for this to be harmless there. Here the scaled direct term is $D_{22} = 0.111$ against a peak $|K_{22}|$ of 1.33: about 8 %, non-negligible but no longer the acute driver it was at a four-times-smaller actuator span, where the same ratio sat on four-times-larger scaled numbers. **The clamp is retained as defence-in-depth rather than as a necessity**: it costs two comparisons per tick, it bounds a failure mode whose severity is a function of the input scaling (and therefore of a number that has already moved once), and the integrator alone should never be able to demand more than the actuator can deliver in any case. The mechanism is exposed only by a closed-loop recovery simulation (rail at -20 A, hold, release: 110 ms on the rail, release within one tick, a 4.0×10^{-4} m s^{-1} recovery tail, no sustained rail). An open-loop "hold a large error then zero it" test cannot reveal it, because a loop with an externally pinned error has no mechanism to unwind.

At this actuator span the 0.5 m s^{-1} drive step no longer saturates at all — it peaks at 11.79 A and settles in 150 ms to 0.500 000 — so the closed-loop recovery simulation above is the only place the $T(0) = I$ property is exercised *through* a saturation episode. It survives it unbiased. The slow tail seen on larger steps is governed by $K_{I,22} = 0.103$ against a 0.122 rad s^{-1} plant pole and is loop shaping, *not* a windup hang.

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